Shareholder Voice and Executive Compensation*

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Abstract

I estimate a model of CEO compensation with nonbinding shareholder approval votes (Say-on-Pay): shareholders can dissent on compensation decisions and punish the Board. Because the compensation proposal is endogenous to the voting environment itself, the threat of dissent disciplines pay ex ante and increases shareholder value by 2.97% on average, despite a 6% failure rate. I analyze counterfactual vote designs that alter the intervention and communication aspects of the vote. Binding votes pin pay to the last approved policy and may ignore new information about CEO ability. Designs that strengthen intervention while incorporating vote-revealed information may best benefit shareholders.

Keywords: shareholder voice, corporate governance, executive compensation, CEO compensation, shareholder voting, say-on-pay, structural estimation, shareholder voting

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1. Introduction

Shareholders elect the Board of Directors but the Board need not represent their interests (Shleifer and Vishny, 1997). A primary manifestation of this problem is in executive compensation: the Board should set compensation to align the interests of management and shareholders, yet directors may have an incentive to favor the CEO (Bebchuk and Fried, 2003).

When shareholders disagree with compensation decisions and exerting control is not viable, shareholders can convey dissent through *voice* (Hirschman, 1970; Cuñat et al., 2016). Say-on-Pay (SOP), a nonbinding shareholder approval vote on CEO compensation policy, is an important channel through which shareholders may voice dissent. While advisory, SOP functions as a public performance evaluation of the compensation committee and a vote of confidence in the Board's pay-setting choices.

Yet the impact of SOP is unclear. Compensation policies receive over 90% support on average and only about 6% of SOP votes in the US fail.[1] This high support may be difficult to reconcile with (i) the literature suggesting that CEOs may influence the pay-setting process (e.g., Bertrand and Mullainathan, 2001; Coles et al., 2014) and (ii) recent survey evidence showing that institutional investors believe CEO pay levels may be too high (Edmans et al., 2023).[2]

An important consideration is that these approval votes are ex post, endogenous outcomes, occurring after compensation has been set and firm performance realized. The pay package brought to a vote is itself an equilibrium object: anticipating how shareholders translate pay and realized performance into dissent, and internalizing the cost of that dissent, the Board may set compensation to make the vote pass. Put differently, the proposal put forth is endogenous to the preferences and technology embedded in the vote itself, and confirming whether low dissent ex post is consistent with effective governance ex ante is an empirical question.

This paper's goal is to pose and estimate a structural model to quantify the influence of SOP on compensation policy and explore the mechanism by which this influence occurs. In the model, the Board decides on compensation policy and may differ from shareholders on the appropriate pay level for the CEO. Shareholders hold the vote: failure may punish the Board, yet dissent from the Board

[1]SOP was formalized in the US as part of Dodd-Frank in 2010. SOP proposals are put forth by management at the annual shareholder meeting, and shareholders vote on the CEO's compensation from the previous fiscal year. I use "failure" to refer to SOP proposals that do not garner the required support from shareholders. SOP votes are nonbinding: there is no threshold which forces the Board to change pay policy. However, the understood threshold for SOP failure, by both academics and practitioners, is 70% support (or 30% voting against, see Dey et al., 2024; ISS, 2022, Section 5 "Compensation"). Further, SOP votes are ex post approval votes on the previous year's CEO compensation, not advisory votes on proposed compensation.

[2]Edmans et al. (2023) survey UK institutional investors; one question they ask is "Do you believe the overall level of CEO pay is too low, too high, or about right?" 77% of investors view pay as too high, and 86% of those agreed that "boards are ineffective at lowering it even though they should." See Section 4.4 and Table OA.4 of the paper.

on a prominent policy may itself be costly to shareholders. Estimating the model will quantify the disciplinary influence of SOP on compensation policy and, more generally, the disciplinary efficacy of shareholder voting, by identifying how proposals are shaped ex ante by the anticipated threat of dissent and whether and how governance frictions make dissent costly, thereby tempering discipline.

However, other factors also influence compensation and voting decisions. The size (or existence) of the difference in preferences between the Board and shareholders over pay levels is not obvious. Some CEOs are more skilled than others and will thus be better compensated. The Board and shareholders cannot observe CEO skill directly and may have different beliefs about the CEO's ability (Taylor, 2010); public and private information will influence learning, and consequently, pay and vote decisions. Separately quantifying the effects of these forces is necessary to fully understand the impact of the vote.

I estimate model parameters via indirect inference. The identification strategy hinges on matching reduced-form regressions that summarize the model's equilibrium interaction between the Board and shareholders: a failure probit that maps pay and performance into dissent, and a wage-setting equation that maps dissent into compensation decisions. The estimated model replicates key moments: it matches the SOP failure rate very closely. It also matches the sensitivity of failure probabilities to both pay and company performance, the primary determinants of SOP vote outcomes (Fisch et al., 2018).

The estimation produces several results. To start, I estimate the magnitude of the difference in preferences over the appropriate level of CEO pay, which I refer to as the "governance wedge" between the Board and shareholders. The wedge implies that Boards behave as if they maximize a weighted average of shareholder profits and CEO utility, with the CEO weight growing over tenure: starting at about 0.24 in the first year of tenure, it rises to roughly 0.29 and 0.34 after 10 and 20 years, respectively. While the magnitude and interpretation of this difference is not the focus of the paper, it implies there is scope for SOP to provide disciplinary value to shareholders.

For the vote to affect compensation policy, however, it must be that the Board internalizes the threat of vote failure into their compensation decisions. To explain observed behavior, I estimate that Boards internalize a utility cost from SOP failure that is equivalent to 2.70% of shareholder value. While the unconditional failure rate of about 6% means the cost is closer to 0.16% of value in expectation, the threat of costly SOP failure disciplines CEO compensation, even if failure may be ex ante unlikely.

Second, failed SOP votes are considered costly by shareholders. I estimate that shareholders in the model internalize a cost to SOP failure equivalent to 1.13% of value (about 0.07% in expectation). This aligns with survey evidence from Edmans et al. (2023): shareholders state that failing the SOP may be undesirable, for example because they are hesitant to dissent from the Board on a prominent policy.

I estimate that providing SOP as a disciplinary mechanism for shareholders lowers pay levels by 7.76% and improves shareholder value by 2.97% on average.[3] These findings are consistent with prominent papers on SOP adoption: Correa and Lel (2016), in a cross-country design on regulatory adoption of SOP, find that the introduction of SOP lowered pay levels by about 7%; and Cuñat et al. (2016), in a close-vote design on US SOP adoption pre-Dodd-Frank, find that it raised firm valuations by about 5%.

The contribution of these findings relative to the adoption-based evidence is the mechanism. In the model, the compensation policy brought to a vote is an equilibrium object: the Board endogenously selects proposals to manage ex ante failure risk. My results show that infrequent dissent is consistent with effective discipline because discipline operates primarily through the threat of dissent. This logic extends beyond SOP. In many shareholder votes on board- or management-sponsored policies, such as director elections (Aggarwal et al., 2019), major transactions (Becht et al., 2016), or equity issuance (Holderness, 2018), the proposal on the ballot (and, in some cases, whether a proposal even comes to a vote) will typically be shaped ex ante by anticipation of shareholder dissent ex post.

While the model presented in this paper captures key institutional features of SOP as a public channel of shareholder voice, it abstracts from two important aspects of the setting. First, I do not model shareholder engagement: private communication that can occur before or after the annual meeting (e.g., Kakhbod et al., 2023; Dey et al., 2024). Engagement is largely unobserved empirically and would require specifying communication-based state variables theoretically. Accordingly, my estimates and counterfactuals should be interpreted as holding the engagement environment fixed; to the extent engagement shapes outcomes in practice, it would do so by altering the effective costs and consequences of public dissent and the degree of information asymmetry between the Board and shareholders.

Second, I abstract from strategic interactions among heterogeneous shareholders, which Barry et al. (2026) studies via an empirical model of strategic shareholder voting. Rather than solve a dynamic voting game with pivotality considerations or strategic coordination, I summarize the shareholder base by a representative voting rule that maps observables into a failure probability. This abstracts from strategic effects by construction and is intended to isolate the paper's main mechanism: how boards internalize the threat of dissent when designing compensation proposals.

Say-on-Pay impacts compensation policy through both a communication and intervention channel. First, the vote outcome is informative: it aggregates shareholder information and conveys it to the Board ex post. Second, dissent is disciplinary: the threat of opposition shapes proposal-setting ex ante.

[3] These magnitudes come from a counterfactual in which I remove the vote from the structural model and re-simulate under the estimated structural parameters (see Section 5.2). As described in the model, compensation affects the CEO's productive input, so changes in pay translate into changes in output, operating income and, ultimately, shareholder value.

Because SOP is advisory, both roles operate indirectly in the baseline model. In the final part of the paper, I use the estimated model to study how vote design should trade off these communication and intervention roles by changing what happens after a failure, by allowing shareholder dissent to directly influence concurrent compensation policy.

In the communication design, the Board is obliged to present a revised compensation package that incorporates vote-revealed information. Relative to the baseline, I find that the communication design has a moderate impact on shareholder value. This moderate impact arises primarily because the counterfactual design improves outcomes in the small set of states in which the vote fails, and because in the baseline model the advisory vote already pushes the Board to account for shareholder views ex ante when setting pay.

In the intervention or binding vote design, the CEO receives the previously-approved pay package, w_{t-1}. This design reduces shareholder value relative to the baseline, first because the realized failure-wage is not conditioned on information about CEO ability arriving since $t-1$. Second, the prospect of that rigidity reduces shareholders' incentives to dissent, which reshapes proposed compensation even along the high-support path.

I also consider a design which mixes aspects of both designs above and makes the failure-wage both partially anchored and partially information-responsive. Perhaps surprisingly, this interior design delivers the largest increase in shareholder value. The reason is that the credible threat of the intervention-based anchor makes dissent consequential enough to tighten incentives ex ante, while preserving scope to incorporate information allows for improvements ex post.

Consistent with Levit and Malenko (2011), the welfare impact of advisory voting hinges on incentives to respond to dissent; here, those incentives are supplied through the baseline advisory vote: making the vote more informative has only an incremental impact. Consistent with Levit (2020), stronger intervention can backfire by reshaping equilibrium behavior, particularly when the failure-induced policy is rigid. One policy implication is that voting mechanisms may best serve shareholders when they make dissent consequential without mechanically locking in an outdated policy; i.e., a credible failure anchor combined with a requirement to revise and resubmit in light of vote-revealed information.

Literature Review. A large empirical literature studies the introduction and consequences of SOP and related pay-vote regulations (e.g., Cai and Walkling, 2011; Ferri and Maber, 2013; Cuñat et al., 2016; Correa and Lel, 2016; Iliev et al., 2015). A central theme is that expanding shareholder voice can reduce pay growth and improve value, but also that realized SOP failures are rare and support rates are high, which complicates inference about the magnitude and channels of SOP's effects (e.g., Armstrong et al., 2013; Kaplan, 2013). My contribution is a structural mechanism: SOP outcomes are endogenous equilibrium objects. The compensation package brought to a vote is chosen anticipating how shareholders translate pay and performance into dissent, so high observed support can be consistent with substantial discipline operating through ex ante incentives rather than frequent ex post failures.

Wagner and Wenk (2025) study policy changes that strengthened binding SOP in Switzerland and document a trade-off in shareholder value across designs; my model provides a mechanism-based decomposition of how making votes more interventionist versus more communicative changes incentives along the equilibrium path, and why hybrid, process-based designs can dominate either extreme.

There is a large literature studying how corporate governance affects firm value. Cuñat et al. (2012) and Flammer (2015) show a causal (positive) relation between adoption of governance-improving mechanisms and firm value; my paper shows how (and by how much) a particular governance mechanism improves shareholder value. Johnson and Swem (2021); Fos (2017) show that, although proxy contests are rare, the threat of initiation influences firm behavior beneficially for shareholders, I show that SOP operates similarly. Holderness (2018) shows that a similar mechanism exists when shareholders vote on equity issuance decisions. Within the realm of hedge fund activism, Gantchev et al. (2018) show that spillover threats of activism induce changes in corporate policies at peer firms; Zhu (2021) focuses on how (threats of) activism influence CEO compensation structures.

This paper relates to the broader literature on shareholder voice (e.g., Hirschman, 1970; Gillan and Starks, 2007), and to work that studies voting as a communication technology whose effectiveness depends on the decision maker's incentives to respond (Levit and Malenko, 2011; Levit, 2019). My results show that recurring votes can matter through equilibrium proposal selection: policies and agendas are shaped ex ante by the anticipated threat of dissent. This logic is not unique to SOP, and connects to evidence that shareholder opposition and governance threats can discipline board- or management-sponsored policies (e.g., director elections, proxy contests, equity issuance, and major transactions, Aggarwal et al., 2019; Fos and Tsoutsoura, 2014; Holderness, 2018; Becht et al., 2016, respectively).

The paper also contributes to the structural executive compensation literature that models CEO pay, learning, and turnover in dynamic settings (e.g., Taylor, 2010, 2013; Page, 2018). I embed a recurring shareholder vote into the pay-setting problem and estimate a reduced-form governance wedge between boards and shareholders that evolves with tenure. This wedge creates scope for shareholder voting/voice to generate value, but whether it does so is an empirical question that depends on the equilibrium consequences of dissent, which my estimation identifies.

A growing structural literature studies shareholder voting. Blonien et al. (2025) study how (the quality) of proxy advisor advice in influencing shareholder voting and find that ISS may lead to errors in shareholder voting. Barry et al. (2026) estimate a model of incomplete information to separate blockholder private information and strategic voting considerations from underlying preferences towards management. My contribution is to estimate a dynamic model in which voting and policy choice are jointly determined and in which both sides have private information, and then study the effectiveness of disciplinary votes (particularly, SOP) on influencing policies.

2. Institutional Background and Data

2.1. Institutional Details of Say-on-Pay

SOP became a standard feature of US corporate governance after the 2010 Dodd-Frank Act mandated that public firms provide shareholders with an advisory vote on executive compensation beginning in 2011 (with some smaller firms phased in later). In practice, nearly all large US firms hold SOP votes annually. The vote is an ex-post, advisory vote: shareholders vote on the compensation outcomes disclosed in the proxy statement for the prior fiscal year, rather than on compensation policy going forward. The vote must cover executive compensation disclosed under Item 402 of Regulation S-K, including the Compensation Discussion and Analysis (CD&A), which is intended to contextualize the Board's rationale for its compensation policy. The model's action and information revelation sequences are chosen to replicate these institutional features.

SOP is advisory, so there is no statutory threshold that forces the compensation committee to revise pay. Instead, what matters is the level of dissent at which key governance intermediaries and boards treat the vote outcome as a negative signal. A central benchmark is 70% support (i.e., 30% votes against). In particular, Institutional Shareholder Services (ISS) treats SOP support below 70% as a low-support outcome that triggers heightened scrutiny of the firm's response, and may lead ISS to recommend votes against the subsequent SOP proposal and/or directors (ISS, 2022, Section 5, "Compensation"). Consistent with this, firms experience sharp changes in institutional investor engagement around the 70% cutoff, while placebo threshold analyses do not show comparable discontinuities at alternative cutoffs such as 50% support (Dey et al., 2024). For these reasons, throughout the paper I define "SOP failure" as at least 30% of votes cast against the SOP proposal.[4]

[4]Moreover, in untabulated analyses, I show that the likelihood of compensation committee director turnover increases significantly in the year after a SOP vote receiving less than 70% support, while there is no increase at the 50% margin. This suggests that, for the entity that makes compensation decisions (the Board of Directors), the 70% margin is more salient.

2.2. Data and Summary Statistics

For the estimation described in Section 4, I merge Say-on-Pay (SOP) vote outcomes from Institutional Shareholder Services (ISS) with CEO compensation from Execucomp and firm-level accounting outcomes from Compustat. The sample covers S&P 1500 firms over 2011–2024. Table 1 reports summary statistics for the key vote, CEO, and firm variables used to construct the targeted moments. SOP votes exhibit high average support: the mean vote against is 8.2% (median 4.6%), and 6.1% of votes exceed 30% against, which is the baseline failure rate used as a moment in estimation. CEO total expected pay (Execucomp, TDC1), the measure of pay used in the estimation, averages \$7.47 million (median \$5.64 million), with sizable within-spell changes (interquartile range for annual pay changes of roughly −10% to +18%). The median length of CEO tenure is seven years, which informs the learning processes in the structural model.

3. Model

3.1. Technology and Environment

Time is annual. The firm is infinitely-lived and consists of three actors: the Board of Directors, which sets the CEO's pay each period; the CEO, who provides a productive input to the firm in return for pay; and a shareholder base, which holds an approval vote over the Board's pay policy. As mentioned above, the wage in the model represents the level of total (expected) pay, with no distinction between base and incentive pay; one interpretation of the model is that the Board and CEO negotiate a pay level each year that determines the CEO input used in production that year (Taylor, 2013).

To link CEO compensation to firm performance in the simplest possible way, I assume a reduced-form relation between the level of pay w_t to the CEO input that enters production:

$$n_t = w_t^{\gamma}.$$

The concavity parameter $\gamma \in (0, 1]$ implies diminishing marginal increases in CEO inputs as pay rises.[5] The firm produces according to

$$y_t = \eta A_t n_t^{\beta},$$

[5] This assumption can be microfounded in several ways. Empirically, w_t is measured by total compensation (TDC1), which bundles salary, bonus, and equity-based pay; to the extent that variation in TDC1 reflects variation in incentive pay, standard principal-agent models with risk-averse agents and convex private costs imply a positive, concave relation between compensation and the CEO input. Internet Appendix IA1.1 provides an alternative microfoundation based on fairness/reference wages (e.g., Chaigneau et al., 2025) that yields the same mapping without explicit pay-for-performance incentives.

where $A_t > 0$ is the firm's productivity and $\beta \in (0, 1)$ governs diminishing returns to the CEO input. Combining with $n_t = w_t^{\gamma}$, output is thus

$$y_t = \eta A_t w_t^{\alpha}, \tag{1}$$

where $\alpha \equiv \gamma\beta \in (0, 1)$ describes the curvature of output in CEO pay and η is a scaling factor, which maps model output into observed firm revenues. These two parameters, to be estimated, control how changes in CEO pay map to changes in firm output.

Firm productivity is centered around CEO skill a but influenced by a mean-zero shock ε_{yt},

$$\ln A_t = a + \varepsilon_{yt}, \quad \varepsilon_{yt} \sim N\left(0, \sigma_y^2\right). \tag{2}$$

Type a is not observed by the Board and shareholders, they make predictions about a based on information they observe (as detailed in Section 3.2). Eq. (2) defines the notion of CEO skill: higher types achieve higher average productivity.

Operating income in year t is

$$\Pi_t = \eta A_t w_t^{\alpha} - \eta\kappa w_t.$$

The parameter $\kappa > 0$ governs how CEO pay translates into firm-level operating costs.[6] The scale parameter η and the cost parameter κ are not essential for the model's strategic forces, but they are important for estimation because they map the model's objects into the revenues and operating income observed in the data. Normalizing operating income by η (so $\pi_t \equiv \Pi_t/\eta$), we have

$$\pi_t = A_t w_t^{\alpha} - \kappa w_t. \tag{3}$$

The Board of Directors (B) sets the wage w_t. A key friction is that the Board's pay-setting objective may place additional weight on CEO pay relative to operating income. Absent dynamic considerations and any influence from Say-on-Pay, the Board chooses w_t to maximize

$$\pi_t^{\mathrm{B}} = A_t w_t^{\alpha} - (1 - \lambda_\tau)\kappa w_t,$$

where $\lambda_\tau \in [0, 1)$ captures the extent to which the Board is willing to trade off operating income for

[6]When $\kappa = 1$ CEO pay and operating costs move in unison; $\kappa \lesseqgtr 1$ implies that the cost impact is smaller or larger, respectively.

higher CEO pay. I allow this to vary across CEO tenure according to

$$\lambda_\tau = \lambda_0(1+\lambda_1)^\tau, \qquad \lambda_\tau \in [0,1). \tag{4}$$

A positive λ_τ can reflect CEO influence on the Board (e.g., board capture, Coles et al., 2014) or, more broadly, level-based pay-setting considerations that the shareholder base does not fully internalize (e.g., reference wages or fairness constraints, Chaigneau et al., 2025).[7] I refer to λ_τ as a "governance wedge," which represents a difference in preferences between the Board and shareholders over the optimal level of compensation.

As Taylor (2013) shows, CEO wages display downward rigidity: risk-averse CEOs accept lower total pay if they are protected from downside risk. To match observed patterns in compensation, the model incorporates an adjustment cost into the Board's compensation decision:

$$AC(w_t; w_{t-1}, \tau_t) = c_w \times w_t \left(\frac{w_t - w_{t-1}}{w_t} \right)^2 \times \mathbb{1}[w_t < w_{t-1}] \times \mathbb{1}[\tau_t > 0]. \tag{5}$$

The adjustment cost is quadratic and only activates if the Board decreases the wage from $t-1$ to t, chosen to match downward rigidity in CEO pay; c_w controls the cost of adjustment, and is to be estimated.

Shareholders hold an approval vote each year on the Board's CEO pay policy, or a Say-on-Pay vote. The vote is nonbinding (failure does not mandate a change in compensation policy) and occurs at the end of each period (after the compensation decision and output have occurred, following the observed institutional structure). A failed vote results in a cost for the Board, $\chi_B \geq 0$. In the model, this cost measures the Board's (perceived) aversion to a failed vote. The Board's per-period utility is

$$A_t w_t^\alpha - \kappa w_t + \lambda_\tau \kappa w_t - \chi_B \times \mathbb{1}[\text{SOP fail}_t] - AC(w_t; w_{t-1}, \tau_t). \tag{6}$$

Shareholders in the model seek to maximize operating income (3): if the proposed wage is high relative to shareholders' beliefs about CEO ability, shareholders may vote against the pay policy. Upon vote failure, shareholders also incur a perceived cost, which can capture (for example) reputational or coordination costs of dissenting from the Board on compensation policy. The parameter $\chi_S \geq 0$

[7]In the model, tenure-based pay growth is attributed to a tenure-increasing Board–shareholder wedge governed by λ_1. An alternative explanation is learning-by-doing: CEO productivity may improve with tenure, raising efficient pay even absent increases in the governance wedge. To the extent such skill growth not fully captured by the model's performance-based updating (see Section 3.2), the model may overstate how much pay growth reflects governance frictions rather than skill accumulation.

governs this cost.[8] Shareholders' per-period utility is

$$A_t w_t^{\alpha} - \kappa w_t - \chi_S \times \mathbb{1}[\text{SOP fail}_t]. \tag{7}$$

Eqs. (6) and (7) summarize the differences between the Board and shareholder objectives. In equilibrium, shareholders can discipline the Board's pay-setting wedge (λ_τ) by threatening to fail the SOP, but this discipline is tempered by the shareholders' own cost of a failed vote. The Board additionally internalizes the adjustment cost.[9]

3.2. Beliefs, Signals and Model Timeline

Each period the firm separates from the CEO with exogenous probability f_τ and matches with a new CEO of tenure $\tau = 0$.[10] At the start of each match, the Board and shareholders begin with the prior about CEO skill,

$$a \sim N\left(\mu_0, \sigma_0^2\right), \tag{8}$$

which matches the distribution of ability in the CEO talent pool. At the annual compensation committee meeting, the Board receives its private signal about the CEO,

$$z_{bt} = a + \varepsilon_{bt}, \quad \varepsilon_{bt} \sim N\left(0, \sigma_{z_b}^2\right). \tag{9}$$

The signal z_{bt} will inform the Board's wage decision, and may represent operational interaction with the CEO. Shareholders do not observe z_{bt}, which means that the Board is asymmetrically informed (more precise beliefs) about CEO skill when setting pay. After pay is set, both output and productivity realize (that is, B does not observe the realization of A_t before setting compensation).

At the annual shareholder meeting, dispersed (atomistic) shareholders form assessments of CEO type using both public information and investor-specific information. These assessments are correlated across shareholders (e.g., through proxy recommendations and common information sources). Rather than modeling an explicit strategic voting game, I summarize the informational content of the shareholder base by a single representative Shareholder, labeled S. Appendix IA1.2 microfounds this

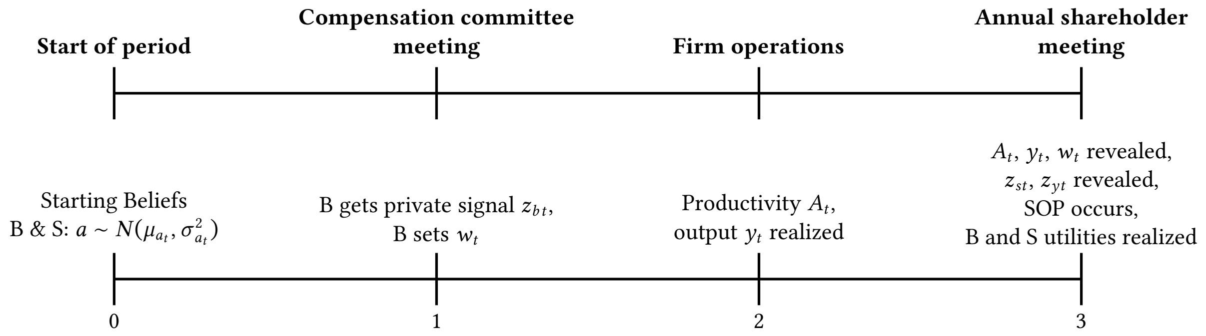
[8]Surveyed institutional investors report reluctance to vote against SOP because they prefer to resolve concerns through engagement rather than public dissent, and because a negative vote can trigger repeated follow-up requests from the firm, "imposing a significant time cost" (Edmans et al., 2023, Online Appendix A).

[9]The assumption that shareholders do not incorporate the adjustment cost helps keeps the shareholders' problem static, which greatly simplifies the numerical solution.

[10]The model takes CEO turnover as exogenous and focuses on compensation and voting within a firm-CEO match. This abstraction allows the analysis to isolate how SOP affects pay policies and Board incentives conditional on a given match.

Figure 1. Model timeline

This figure displays the within-period model timeline. The top timeline displays the timeline as it maps to practice; the bottom as it maps to the sequencing of events within each period t. Figure IA.1 displays an in-depth timeline which incorporates the timing of the strategies by the Board and Shareholder (See Appendix IA1.5).



reduction by showing that, under correlated information and atomistic ownership, the vote outcome can be represented using a sufficient statistic for the shareholder base's information.

At the end of each period t, S observes an aggregate private signal about CEO type,

$$z_{st} = a + \varepsilon_{st}, \quad \varepsilon_{st} \sim N\left(0, \sigma^2_{z_s}\right). \tag{10}$$

S also observes output y_t and the CEO's pay w_t. These observables imply a public signal about productivity A_t, which is informative of the CEO's type. I label this signal

$$z_{yt} = \ln A_t = a + \varepsilon_{yt}, \quad \varepsilon_{yt} \sim N\left(0, \sigma^2_{y}\right). \tag{11}$$

Both the private signal z_{st} and the public signal z_{yt} will affect the SOP result.

Model timeline. Figure 1 displays the per-period sequence of events. I assume signal disclosure in the wage-vote environment: given the equilibrium mapping from signals to actions, the Board's wage choice and the SOP vote outcome reveal the underlying signals z_{bt} and z_{st}. As a result, at the beginning of each period t, B and S share a common belief about CEO ability, $a \sim N(\mu_{at}, \sigma^2_{at})$, see Appendix IA1.5. At the compensation committee, B receives its signal z_{bt}, which informs their wage decision. Then, operations take place: the CEO supplies their input, productivity realizes and output occurs. At the annual shareholder meeting, S receives the private signal z_{st}, and output, productivity and wages are revealed; productivity reveals the public signal z_{yt}. Finally, the SOP vote occurs.

Board and Shareholder beliefs. Both B and S use Bayes' rule to update beliefs about CEO ability after their signals. I use subscript a to refer to beliefs shared by B and S: $\left(\mu_{at}, \sigma^2_{at}\right)$ refers to the shared prior at the beginning t. Subscripts b and s refer to when B and S can have different beliefs. CEO tenure fully determines the variance of beliefs; the function $\sigma_a^2(\tau)$ tracks how the variances of beliefs decreases across tenure. Given tenure τ_t at t,

$$\sigma_{at}^2 = \sigma_a^2(\tau_t) = \sigma_0^2\left[1 + (\tau_t) \times \sigma_0^2\left(\sigma_{z_b}^{-2} + \sigma_{z_s}^{-2} + \sigma_y^{-2}\right)\right]^{-1}, \quad \sigma_a^2(0) = \sigma_0^2. \tag{12}$$

The mean evolves according to

$$\mu_{at+1} = \sigma_a^2(\tau_t + 1)\left[\frac{\mu_{at}}{\sigma_a^2(\tau_t)} + \frac{z_{bt}}{\sigma_{z_b}^2} + \frac{z_{st}}{\sigma_{z_s}^2} + \frac{z_{yt}}{\sigma_y^2}\right], \tag{13}$$

and the rate of decline of the variance σ_{at}^2 means that μ_{at} tends toward the CEO's true ability.

3.3. The Say-On-Pay Vote

The Board and Shareholder's pay and vote decisions are informed by each party's private signal and how they view each other's beliefs. This section details the Shareholder's strategy, holding the Board's wage choice fixed. To keep the vote tractable and focus on its role in disciplining pay, I make two broad assumptions. First, I model S' voting strategy as a threshold rule. This rule depends on the information available at the annual meeting and the observed wage. Importantly, I treat the threshold rule as selected ex ante (before within-period private signals are realized); this lets SOP affect the Board's wage choice through an ex ante probability of failure, even though a realized failure is costly. This can be interpreted as reduced-form commitment to voting guidelines (e.g., proxy-advisor and institutional-voting policies) or as a credibility device needed for SOP to discipline the Board. Second, S votes only based on contemporaneous pay and signals, rather than solving a dynamic voting problem. This avoids introducing an additional shareholder value function into the solution, substantially reducing the computational cost of the numerical solution.[11]

3.3.1. The Vote Decision

Informally, S fails the SOP when the CEO's wage is "too high" given S' beliefs about CEO type and the cost of vote failure. The notion of "too high" is summarized by a threshold rule: the Shareholder chooses an upper bound on the acceptable wage, and fails if the Board's wage exceeds this bound.

[11] In untabulated reduced-form results, SOP voting outcomes respond primarily to contemporaneous pay, performance, and revenues, with limited sensitivity to lagged measures.

Formally, conditional on the wage choice, S' strategy can be represented as the threshold:

$$k_{st} = s_t \times w_t, \tag{14}$$

where s_t is the Shareholder's choice variable. After observing signals at the shareholder meeting, they compute a shareholder-optimal benchmark wage implied by their posterior, which I label w_t^{S} and define below. The vote fails whenever the benchmark falls below the threshold:

$$\text{SOP fail}_t \qquad \Longleftrightarrow \qquad w_t^{\mathrm{S}} \leq k_{st} = s_t \times w_t.$$

Thus, s_t controls the sensitivity of failure probability to changes in the Board's wage choice w_t: a lower s_t makes shareholders more tolerant of deviations from their benchmark.

At the annual shareholder meeting, the Shareholder receives its private signal z_{st} and the public productivity signal $z_{yt} = \ln A_t$. The Shareholder combines these into a signal about CEO type, which has distribution

$$Z_{syt} = pz_{st} + (1-p)z_{yt} = a + \varepsilon_{syt} \quad Z_{syt} \mid \left(\mu_{at}, \sigma_{at}^2\right) \sim N\left(\mu_{at},\, \sigma_{at}^2 + \sigma_{sy}^2\right), \tag{15}$$

where ε_{syt} precision-weights the private and public shocks and has variance $\sigma_{sy}^2 = \frac{\sigma_{z_s}^2 \sigma_y^2}{\sigma_{z_s}^2 + \sigma_y^2}$.

Distance from the shareholder-optimal benchmark w_t^{S}. The SOP rule is naturally indexed by the gap between the observed wage and shareholder-optimal wage implied by S' information after the meeting. After seeing meeting-based information Z_{syt} and updating beliefs, the shareholder-optimal wage w_t^{S} would arise out of the following program:

$$\arg\max_{w_t} \quad \mathbb{E}_{st}\left[\exp\left(a + \varepsilon_{yt}\right)\right] w_t^{\alpha} - \kappa w_t = \left(\frac{\alpha}{\kappa}\mathbb{E}_{st}[\exp\left(a + \varepsilon_{yt}\right)]\right)^{\frac{1}{1-\alpha}}.$$

Ex ante, at the time S chooses its strategy, w_t^{S} has the following distribution, given prior $(\mu_{at}, \sigma_{at}^2)$:

$$w_t^{\mathrm{S}} \mid \left(\mu_{at}, \sigma_{at}^2\right) \sim \log N\left(m_t^S, v_t^S\right), \quad m_t^S \equiv \frac{\mu_{at} + \log\frac{\alpha}{\kappa}}{1-\alpha} + \frac{1}{2}\frac{\sigma_{st}^2 + \sigma_y^2}{(1-\alpha)^2}, \quad v_t^S = \frac{\sigma_{at}^2 - \sigma_{st}^2}{(1-\alpha)^2} \tag{16}$$

where $\sigma_{st}^2 = \frac{\sigma_{at}^2 \sigma_{sy}^2}{\sigma_{at}^2 + \sigma_{sy}^2}$ is the Shareholder's posterior variance. I refer to the CDF of this distribution as

F_t^{S}, and use it to assess the probability of vote failure in the Board and Shareholder's objectives:

$$w_t^{\mathrm{S}} \mid \left(\mu_{at}, \sigma_{at}^2\right) \sim F_t^{\mathrm{S}}.$$

3.3.2. Determining the Probability of Say-on-Pay Failure

Given the threshold rule in (14), the SOP vote fails when the information available at the annual meeting implies that the CEO wage is high relative to S' benchmark wage. Thus, conditional on the Board's wage choice w_t, the ex ante probability of failure is

$$\Pr(\text{SOP fail}_t) = \Pr\left(w_t^{\mathrm{S}} \leq s_t \times w_t\right) = F_t^{\mathrm{S}}(s_t \times w_t). \tag{17}$$

As discussed above, SOP is modeled as an ex ante rule that disciplines the Board through the anticipated probability of failure. Accordingly, S chooses the policy parameter s_t ex ante, anticipating the effects that meeting-based information will have on beliefs. Importantly, because the Board observes z_{bt} before choosing w_t, the Shareholder must set their threshold in expectation over the Board's information (i.e., S integrates over the distribution of z_{bt} when choosing s_t), and the Board subsequently chooses $w_t(z_{bt}, s_t)$ given its signal and S' choice. The Shareholder's problem is therefore

$$\max_{s_t} \quad \int_{z_b} f(z_b) \left[\mathbb{E}_{st}[A_t] w_t(z_{bt}, s_t)^{\alpha} - \kappa w_t(z_{bt}, s_t) - \chi_S \times F_t^{\mathrm{S}}(s_t \times w_t(z_{bt}, s_t))\right] dz_b. \tag{18}$$

Commitment to a costly failed vote. A key feature of the model is that the Shareholder choose a voting policy s_t ex ante (as a function of the information state) and then implements the threshold rule at the meeting, even though a realized failure is costly. This commitment is essential for the vote to have disciplining power: absent commitment, shareholders could choose a policy that induces a positive failure probability to influence w_t and then renege to avoid paying the failure cost.

In practice, this reduced-form commitment captures two institutional features of US voting. First, voting is repeated and relationship-based: boards anticipate that today's SOP outcome affects future engagement, director support, and the credibility of shareholder dissent, so reneging today weakens the threat that disciplines future pay. Second, large institutional investors often follow pre-specified stewardship and voting guidelines; deviating ex post from an announced or internally-adopted voting rule can be reputationally costly and can reduce future influence.

Appendix IA1.3 provides a simple relational microfoundation (in the spirit of Bull, 1987) in which failing a vote is a self-enforcing punishment: after observing a sufficiently high wage, shareholders prefer to incur a one-time cost today in order to preserve a lower-wage continuation path in the future. The key condition is that the present value of future discipline exceeds the contemporaneous cost.[12]

3.4. The Compensation Decision

Each period, the Board observes its private signal z_{bt} and chooses the CEO's wage w_t, taking as given the Shareholder's SOP policy s_t. At the start of t, the common belief about CEO type is $a \sim N(\mu_{at}, \sigma^2_{at})$. Upon observing z_{bt}, the Board updates its posterior to $\left(\mu_{bt|z_b}, \sigma^2_{bt|z_b}\right)$, where

$$\begin{aligned} \mu_{bt|z_b} &= \sigma^2_{bt|z_b}\left(\frac{\mu_{at}}{\sigma^2_{at}} + \frac{z_{bt}}{\sigma^2_{z_b}}\right) \\ \sigma^2_{bt|z_b} &= \left(\sigma^{-2}_{at} + \sigma^{-2}_{z_b}\right)^{-1} = \frac{\sigma^2_{at}\sigma^2_{z_b}}{\sigma^2_{at} + \sigma^2_{z_b}}. \end{aligned} \tag{19}$$

The Board's wage policy solves the following Bellman equation:

$$\begin{aligned} V(\mu_{at}, \tau_t, w_{t-1}) = \max_{w_t} \quad & \mathbb{E}_{bt|zb}[A_t]w_t^{\alpha} - (1-\lambda_\tau)\kappa w_t - \chi_B \times F^S_t(s_t \times w_t) + AC(w_t, w_{t-1}; \tau_t) \\ & + \delta_B\Big[(1-f_{\tau_t})\mathbb{E}_{bt|zb}[V(\mu_{at+1}, \tau_t+1, w_t)] + f_{\tau_t}V^R\Big]. \end{aligned} \tag{20}$$

The state consists of the current shared belief mean μ_{at}, CEO tenure τ_t (which pins down belief dispersion; see Eq. 12), and the previous period's wage w_{t-1} (set to zero when $\tau_t = 0$). The term $F^S_t(s_t \times w_t)$ is the (anticipated) probability that SOP fails, as detailed in Section 3.3. The expectation operator $\mathbb{E}_{bt|z_b}[\cdot]$ is taken with respect to the Board's posterior after observing z_{bt}, i.e., using (19). The hazard function f_{τ_t} governs CEO-firm separation and is taken as an input to the model.[13] Upon separation, the Board hires a new CEO and beliefs reset to (μ_0, σ_0), so the termination value is

$$V^R = V(\mu_0, \tau = 0, 0). \tag{21}$$

[12]Indeed, as we show in Section 5.2, removing SOP as a disciplining device would lower shareholder value by about 3%. While the commitment mechanism is modeled in reduced form, the estimated economy implies that shareholders would strictly prefer a regime with a credible SOP punishment technology: the value loss from removing SOP is large, whereas the equilibrium incidence of costly failures is low, so the expected utility cost of occasionally failing a vote is small relative to the disciplining benefits.

[13]Pooling all CEO spells, f_τ is the frequency of turnover after τ years, conditional on surviving $\tau - 1$ years. I follow Taylor (2010) in computing these hazard rates. For tractability, turnover is treated as exogenous; the analysis focuses on pay-setting and SOP discipline within an ongoing CEO-firm match. For simplicity, $f_0 = 0$ in estimation, and $f_\tau = 1$ at the tenure cap $\mathbf{T}$. The estimation sample excludes CEO spells that last only one year.

3.4.1. Shareholder Value

Given the Board's equilibrium wage policy $\omega_t = \omega(\mu_{at}, \tau_t, w_{t-1})$, I evaluate outcomes using a shareholder value criterion: the discounted present value of operating income implied by the equilibrium policy,

$$V^{\mathrm{SH}}(\mu_{at}, \tau_t, w_{t-1}) = \mathbb{E}_{at}[A_t]\omega_t^{\alpha} - \kappa\omega_t + \delta_B\Big[(1 - f_{\tau_t})\mathbb{E}_{at}\big[V^{\mathrm{SH}}(\mu_{at+1}, \tau_t + 1, \omega_t)\big] + f_{\tau_t}V^{\mathrm{SH},R}\Big], \tag{22}$$

with $V^{\mathrm{SH},R} \equiv V^{\mathrm{SH}}(\mu_0, \tau = 0, 0)$. This differs from the Board's objective because it values only operating income cash flows: the pay-setting wedge λ_τ and the perceived vote and adjustment costs only enter via the Board's pay decision, and SOP enters only through its impact on the equilibrium wage policy ω_t. I use (22) to evaluate counterfactual changes to the SOP environment.

3.5. Model Solution

I apply Bayes' rule to characterize the Board's and the Shareholder's posterior beliefs about CEO type and substitute these beliefs into the Board's Bellman equation (20) and the Shareholder's objective (18). I compute a Markov equilibrium in stationary policies as follows. For each Board state (μ_t, τ_t, w_{t-1}), I start from a candidate Shareholder SOP policy parameter s_t. Taking this s_t as given, I solve the Board's dynamic program to obtain the wage policy $\omega(\mu_t, \tau_t, w_{t-1})$. Given the induced wage policy, I then update the Shareholder's best response for s_t. I iterate between these two steps until the wage and vote policies converge. Appendix IA1.6 provides full computational details.

4. Estimation

I estimate the parameters of the structural model using indirect inference (McFadden, 1989): I choose the vector of structural parameters which minimizes the difference between the reduced-form outcomes of an auxiliary model estimated on observed and simulated data. As is standard, I weight moments by the inverse of the estimated variance-covariance matrix of the data moments, clustered by firm–CEO spell. Further details of the estimation are presented in Appendix IA2.

4.1. Identification Strategy

I show there is a tight relation between the reduced-form moments of the auxiliary model and structural parameters. Table A.1 displays the reduced-form outcomes and the parameter(s) each outcome targets.

4.1.1. Output and CEO skill parameters

Because the structural model has neither fixed firm or time heterogeneity, I purge these components from observed revenues using a longer panel (1992–2024) by estimating the auxiliary FE decomposition

$$\log \text{rev}_{fit} = c_{\text{rev}} + \psi_f + \phi_i + \lambda_t + u_{fit},$$

where f, i and t indicate firm, CEO-spell and year, respectively. I then construct the empirical counterpart to the model's definition of log revenues as

$$\log y_{it}^{\text{data}} \equiv \log \text{rev}_{fit} - \hat{\psi}_f - \hat{\lambda}_t.$$

In the structural model,

$$\log y_{it} = \log \eta + a_i + \alpha \log w_{it} + \varepsilon_{yit}.$$

Since $\mathbb{E}[a_i]$ and $\log \eta$ are not separately identified, I normalize $\mu_0 = \mathbb{E}[a_i] = 0$ and target $\mathbb{E}[\log y_{it}^{\text{data}}]$ to pin down $\log \eta$. To discipline $(\alpha, \sigma_y, \sigma_0)$ I use an auxiliary regression on the estimation sample (2011–2024):

$$\log y_{it}^{\text{data}} = y_1 \log w_{it} + \mu_i^y + \epsilon_{it}^y. \tag{23}$$

I target the auxiliary moments $\hat{y}_1$, $\text{Var}(\hat{\epsilon}_{it}^y)$, and $\text{Var}(\hat{\mu}_i^y)$.[14] $\hat{y}_1$ is treated as a reduced-form elasticity, and helps identify α. $\text{Var}(\hat{\epsilon}_{it}^y)$ identifies σ_y and $\text{Var}(\hat{\mu}_i^y)$ identifies σ_0.

In the model, operating profits are

$$\Pi_{it} = y_{it} - \kappa \eta w_{it}, \qquad y_{it} = \eta A_{it} w_{it}^{\alpha},$$

so the model-implied cost share satisfies the identity $1 - \Pi_{it}/y_{it} = \kappa(\eta w_{it}/y_{it})$. In the data, I define operating costs net of CEO pay:

$$s_{it}^c \equiv \frac{\text{Sale}_{ft} - \text{OIBDP}_{ft} - w_{it}}{\text{Sale}_{ft}}, \qquad s_{it}^w \equiv \frac{w_{it}}{\text{Sale}_{ft}},$$

[14]To ensure consistency between data and model, I compute these reduced-form objects using the same fixed-effect and residualization procedures in both observed and simulated data, so that any finite-sample estimation noise in μ_i^y and $\hat{y}_1$ is mirrored in the simulated moments.

and estimate the auxiliary regression

$$s_{it}^{c} = k_0 + k_1 s_{it}^{w} + \epsilon_{it}^{k}. \tag{24}$$

I target the reduced-form slope $\hat{k}_1$ to identify the structural scale parameter κ.

4.1.2. The Board-Shareholder Game

Identifying the parameters relating to the wage and vote decisions is a crucial empirical step, and the identification argument centers on matching equations that describe (in reduced-form) the model's equilibrium interaction between the Board and Shareholder.

The probability of SOP failure. The model's definition of SOP failure is

$$\mathbb{1}[\text{SOP fail}_{it}] = \mathbb{1}\left[s_{it} w_{it} - w_t^{S} \geq 0\right],$$

where w_t^S is the benchmark wage induced by the Shareholder's posterior post-meeting (see Section 3.3.1), and s_{it} is the Shareholder's choice variable. This implies a probability of failure (conditional on information revealed up to period t, $\mathcal{I}_{it}$) equal to

$$\Pr(\text{SOP fail}_{it} \mid \mathcal{I}_{it}) = \Phi\left(\frac{\log(s_{it} w_{it}) - c_0 - c_1 \mu_{it}}{c_1 \sigma_{z_s}}\right), \tag{25}$$

where (c_0, c_1) are constants that map (posterior) belief μ_{it} to the shareholder-optimal wage distribution.

First, the unconditional failure rate is informative about the Shareholder's aversion to failure χ_S: a higher χ_S will lead to a lower choice of s_{it} on average and a lower rate of failure across the entire sample. The model-implied determination of SOP failure maps to the probit regression

$$\Pr\left(\text{SOP fail}_{it} \mid w_{it}, \epsilon_{it}^{y}, \mu_i^{y}\right) = \Phi\left(s_0 + s_1 \log w_{it} + s_2 \epsilon_{it}^{y} + s_3 \mu_i^{y}\right) \tag{26}$$

where ϵ_{it}^{y} is the residual from the output regression (23), and μ_i^{y} is the estimated CEO fixed effect.[15]

In the estimation, I target the average marginal effects (AME) implied by s_1, s_2, and s_3.[16] The

[15]Two equilibrium implications motivate the auxiliary probit. First, under Gaussian signals, posterior beliefs are affine in a persistent ability component, output innovations, and the shareholder's private signal. Second, the shareholder's choice s_{it} is an equilibrium object measurable with respect to $\mathcal{I}_{it}$, so $\log(s_{it} w_{it})$ is also an affine function of the same state variables (up to an approximation error). This yields a probit index that is linear in compensation, output innovations, and persistent CEO heterogeneity, with z_{sit} absorbed into the latent error.

[16]I report AMEs as the percentage-point change in failure probability induced by a 1% change in x (e.g., a change in log compensation); the scaling is for interpretation only.

AME of log compensation on SOP failure (conditional on ability) is highly informative about both χ_S and χ_B: $\hat{s}_1$ is decreasing in χ_S, which dampens the sensitivity of failure to the wage, and increasing in χ_B, as the Shareholder internalizes that SOP failure has a larger impact on compensation (all else equal). Because the shareholder adjudges failure based on their posterior belief about CEO ability, including the output effect gives a failure-wage sensitivity that is conditional on CEO ability. Moreover, as shareholders learn about CEO ability via output innovations, including the estimated output residual isolates residual variation as attributable to private signal realizations (which we discuss further below).

The Board's compensation decision. The Board's pay choice depends on the Board's beliefs about CEO productivity, the size of the governance wedge, the probability of SOP failure, and an adjustment cost if the Board lowers pay. To start, consider a static version of the Board's problem (i.e., we abstract away from dynamics and the adjustment cost).[17] The resulting first-order condition yields the following first-order approximation of the static Board-optimal log wage level

$$(1-\alpha)\log w_{it} \approx \log\frac{\alpha}{\kappa} + \log \mathbb{E}_{bt}[A_{it}] - \log(1-\lambda_{\tau_t}) - \frac{\chi_B}{\kappa(1-\lambda_{\tau_t})}\frac{\partial \Pr(\text{SOP fail}_{it} \mid \mathcal{I}_{it})}{\partial w_{it}}. \tag{27}$$

The Board's decision is driven by three terms: beliefs about CEO ability, a tenure-specific term, and how the Board internalizes (changes in) Pr(SOP fail) from increasing pay.

The following regression provides a reduced-form description of the Board's decision

$$\log w_{it} = b_1 \mathbb{1}[\text{SOP fail}_{it}] + b_2\tau_{it} + \mu_i^b + \epsilon_{it}^b, \tag{28}$$

where μ_i^b is a CEO fixed effect and τ_{it} is the CEO's tenure. I target the average log compensation, which helps most in identifying λ_0. Because the skill parameters are identified, λ_0 is identified by matching average wages. The coefficient on tenure, $\hat{b}_2$ helps most in identifying λ_1: conditional on the vote result and the CEO effect, it captures growth in compensation across tenure independent of variation in (beliefs about) ability.

The coefficient on the SOP failure indicator $\hat{b}_1$ is primarily informative about the Board cost χ_B. From (27), χ_B scales the marginal discipline term: holding fixed shareholders' voting rule, a higher χ_B induces the Board to lower compensation more strongly in states with greater failure risk.[18] I also

[17]The static program is

$$\max_{w_{it}} \quad \mathbb{E}_{bt}[A_{it}]w_{it}^{\alpha} - \kappa(1-\lambda_{\tau_{it}})w_{it} - \chi_B \Pr(\text{SOP fail}_{it}).$$

The expression in the text results from taking first-order conditions, re-expressing the optimal w_{it} in log terms and making the (approximate) substitution $\log(1+x) \approx x$ for small $\frac{\chi_B}{\kappa(1-\lambda_{\tau_t})}\frac{\partial \Pr(\text{SOP fail}_{it})}{\partial w_{it}}$.

[18]Using the realized indicator is appropriate for indirect inference because it is an equilibrium outcome whose conditional

match the variance of the CEO fixed effect, which provides information about variability in both CEO ability and Board beliefs (i.e., σ_{z_b}). Lastly, the log wage autocovariance helps identify c_w:

$$\text{Cov}(\log w_{it}, \log w_{it-1}). \tag{29}$$

Joint identification of χ_S and χ_B. These costs are separated using the two auxiliary equations (26) and (28). The shareholder probit pins down the reduced-form relation between observables and Pr(SOP fail), in particular the slope of failure with respect to compensation and the unconditional failure rate; these moments are primarily informative about the shareholder's aversion to failure χ_S because χ_S shifts the level and sensitivity of failure probability. Holding this fixed, the Board's wage equation then identifies χ_B from the effect of failure on ex ante pay: χ_B scales the discipline term in (27) and therefore governs how strongly the Board adjusts pay when failure risk increases.

Identifying private information. The model includes both Board and Shareholder private information, which are central to realized pay and vote decisions.

Shareholder private information. Appendix IA1.2 shows that with a dispersed shareholder base who receive correlated private signals, the realized proportion of shareholders voting against is informationally equivalent to the realization of a representative shareholder's signal. An implication is that the vote share in excess of the ex ante probability of failure directly reflects (aggregate) private shareholder information. The arguments in Appendix IA1.2 establish a correspondence between the observed vote proportion (in excess of predicted failure probabilities) and shareholder private information.

Letting N_{it} denote the latent index from the probit in (26), the ex ante probability of SOP failure is $\Phi(N_{it})$ and the realized vote share against can be written as

$$V_{it}^{\text{against}} = \Phi(N_{it} + \tilde{z}_{sit}),$$

where $\tilde{z}_{sit}$ captures innovations in (aggregate) shareholder private information that affect vote outcomes. The excess dissent relative to the ex ante probability of failure,

$$\tilde{V}_{it}^{\text{against}} \equiv V_{it}^{\text{against}} - \Phi(N_{it}) = \Phi(N_{it} + \tilde{z}_{sit}) - \Phi(N_{it}),$$

expectation equals the ex ante failure probability given $\mathcal{I}it$; since w_{it} is chosen with respect to $\mathcal{I}_{it}$, the wage-failure covariance, and hence the projection coefficient $\hat{b}_1$, summarizes how compensation co-moves with the threat of SOP failure. This moment therefore helps to identify χ_B.

is highly informative about the σ_{z_s}. Intuitively, the model states that, conditional on the information at time t (and, particularly, the output residual, from the econometrician's perspective), only private signal realizations drive variation in the realized proportion voting against relative to the ex-ante probability of failure. The variance of this difference is proportional to the variance of the Shareholder's aggregate private information. In the estimation, I target the cross-sectional variance $\text{Var}(\tilde{V}_{it}^{\text{against}})$, which is increasing in $\sigma_{z_s}^2$.[19]

Board private information. Under the learning structure in the model, the posterior variance of CEO ability after τ periods is

$$\sigma_a^2(\tau) = \left(\sigma_0^{-2} + \tau \times K\right)^{-1}, \qquad K \equiv \sigma_y^{-2} + \sigma_{z_s}^{-2} + \sigma_{z_b}^{-2},$$

The posterior mean after τ periods, $\mu_{i\tau} = \mathbb{E}[a_i \mid \tau]$, has variance

$$Var\left(\mu_{i,\tau}\right) = \sigma_0^2 - \sigma_a^2(\tau).$$

In other words, types separate across tenure. While ability as a function of tenure is not directly observable, a natural proxy is the CEO fixed effect in the Board equation (28). Importantly, variation in the estimated fixed effect is already conditional on tenure, so provides separate identifying variation in CEO ability within-tenure. Intuitively, variation in CEO spell lengths across tenure pin down how information causes the cross-sectional belief about CEO ability to separate (i.e., drive up the variance of the conditional mean as tenure increases). I estimate the following slope relation:

$$\left(\widehat{\mu_i^b} - \overline{\mu_i^b(\tau_{it})}\right)^2 = v_1 \tau_{it} + v_{it} \tag{30}$$

The dependent variable measures how far i's fixed effect lies from the average effect at tenure τ_{it}. Thus, $\hat{v}_1$ helps identify Board learning across tenure, and because $\left(\sigma_0^2, \sigma_y^2, \sigma_{z_s}^2\right)$ are identified, $\hat{v}_1$ pins down $\sigma_{z_b}^2$.

4.1.3. Correlations between Targeted Moments

Table A.1 Panel B reports pairwise correlations across the targeted moments. The moments that discipline the Board and shareholder costs come from distinct conditional projections and are not collinear.

[19] A second-order expansion around $\tilde{z}_{sit} = 0$ yields $\Phi(N_{it} + \tilde{z}_{sit}) - \Phi(N_{it}) = \phi(N_{it})\tilde{z}_{sit} - \frac{1}{2}N_{it}\phi(N_{it})\tilde{z}_{sit}^2 + \mathcal{O}(\tilde{z}_{sit}^3)$. Consequently, $\text{Var}(\tilde{V}_{it}^{\text{against}} \mid N_{it}) = \phi(N_{it})^2\sigma_{z_s}^2 + \mathcal{O}(\sigma_{z_s}^4)$, and hence $\text{Var}(\tilde{V}_{it}^{\text{against}})$ is proportional to $\sigma_{z_s}^2$ up to the factor $\mathbb{E}[\phi(N_{it})^2]$.

Although higher pay predicts a higher probability of SOP failure in the shareholder probit, the unconditional failure component (s_0, column 6) is only moderately correlated with average log compensation (column 10; corr $\approx$ 0.17). The Board's pay response to failure (b_1, column 11) is essentially orthogonal to s_0 (corr $\approx$ 0.05) and to average pay (corr ≈ -0.03), consistent with b_1 capturing state-contingent pay compression rather than a level effect.

Most importantly for separating χ_S and χ_B, the SOP failure-wage sensitivity (s_1, column 7) and the pay gap across failure states (b_1, column 11) are not highly correlated (corr $\approx$ 0.40). As argued above, (26) disciplines how failure likelihood varies by wages and performance, and loads primarily on χ_S, while (28) disciplines the Board's state-contingent wage response and loads primarily on χ_B. These moments are not redundant and provide the necessary variation for jointly identifying (χ_S, χ_B).

4.2. Estimated Model Fit

4.2.1. Moment-Matching Exercise

Table 2 displays the closeness of auxiliary-model moments from the the observed and simulated data.

Output & CEO skill moments. Both scale moments (average output and the cost share moment) are well-matched. The model misses the elasticity of output to the wage level (moment 3, y_1: 0.284 in the observed vs. 0.158 in the simulated data). To match moments in the Board-shareholder game, the estimation requires a large σ_{z_b} relative to σ_0. This creates an attenuation bias in the simulated data, which leads to the low value of y_1.[20]

The estimation matches the variance of the CEO fixed effect in output (moment 4) very closely, suggesting that σ_0 is well-identified. With respect to the variance of the output residual, the estimation requires a large σ_y. This suggests there are issues in separately identifying output innovations from CEO effects in the data. If any volatility parameters are smaller than the prior volatility σ_0, learning about CEO ability trends toward the true value very quickly. As such, to match rates of updating in the Board and shareholder moments, the model pushes σ_y up.

Shareholder moments. The failure rate (moment 6) is matched closely: 6.1% and 5.9% in the observed and simulated data, respectively. The sensitivity of Pr(SOP fail) to the wage is also well-matched (moment 7, s_1). These are key moments for identifying shareholder preferences concerning

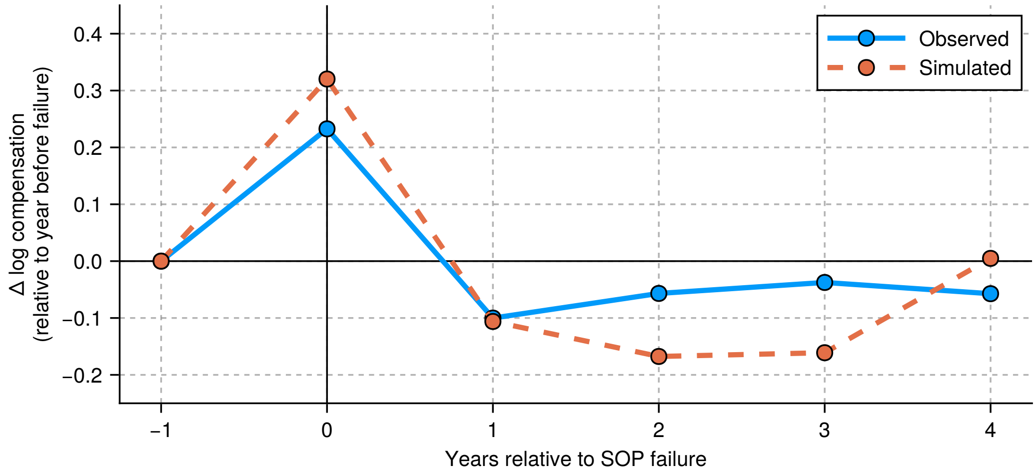
[20]In the estimated model, matching the Board–shareholder strategic moments requires the Board's private signal to be relatively imprecise (a large σ_{z_b} relative to the prior dispersion σ_0). As a result, wages are noisier than the underlying ability component that drives A_t. This reduces the share of wage variation that is informative about differences in productivity, so the coefficient from the simulated data (net of CEO effects) is below α.

Figure 2. Model fit: Compensation dynamics around SOP failure

This figure displays compensation dynamics around SOP failure in both the observed and simulated data. On both datasets, I run the following difference-in-difference regression:

$$\log w_{it} = \iota_i + \sum_{k=-1}^{4} \beta_k \mathbf{1}\left\{ \left(t - T_{it}^{fail}\right) = k \right\} + \gamma \cdot X_{it} + \varepsilon_{it},$$

where i indicates CEO spell, t indicates (calendar) time, ι_i is a CEO-spell fixed effect and T_{it}^{fail} indicates the time of the SOP failure. In the observed data regression, I include calendar time fixed effects to account for aggregate shocks/trends; in both regressions I include a control for that period's output shock. Each point on the plot displays the predicted difference in log compensation at time k relative to $k = -1$ (year before failure).



vote outcomes, so the closeness is reassuring.

The sensitivity of Pr(SOP fail) to output innovations and predicted CEO output effects match sign in the simulated data, but are large in magnitude relative to the observed data. The model's parsimony may ascribe too much predictive power to these moments. Lastly, the excess vote against variance (moment 10), is closely matched, suggesting that σ_{z_s} is identified.

Board moments. Overall, the Board moments are matched closely. Average log compensation (moment 11) is 1.659 in the data versus 1.696 in simulation, indicating that baseline pay levels, and thus baseline capture λ_0 conditional on the output block, are well disciplined. The within-CEO pay gap across SOP outcomes (moment 12, b_1) is also close (0.334 observed versus 0.386 simulated). This gap is a key statistic for the Board's discipline cost χ_B.

The tenure slope of compensation (moment 13, b_2) is well matched (0.038 observed versus 0.033 simulated), supporting identification of the tenure profile of the governance wedge (e.g., λ_1). Lastly, the autocovariance of log pay (moment 15) is somewhat higher in simulation (0.650) than in the data (0.594), indicating that the model generates slightly too much wage persistence. The cross-sectional variance of the CEO fixed effect in the Board equation (moment 14, 0.571 observed versus 0.553 simulated), and the tenure-gradient dispersion moment (moment 16, 0.008 versus 0.007) are both closely matched, suggesting that the model captures heterogeneity in Board beliefs.

4.2.2. Parameter–Moment Sensitivity

I assess local identification using the moment-sensitivity diagnostic in Andrews et al. (2017). Table A.2 shows standardized sensitivities (ranging from -1 to 1). The Board's rejection cost χ_B is primarily disciplined by b_1, the pay gap across failure states in the Board regression (28). By contrast, the Shareholder's failure cost χ_S is pinned down by the moments from the probit (26), most directly s_1 (the AME of $\log w_{it}$) and s_2 (the AME of output innovations ϵ^y_{it}). The two private-signal noise parameters are also separately identified. The Shareholder signal noise σ_{z_s} is disciplined by $\mathrm{Var}\big(\tilde{V}^{\text{against}}\big)$. The Board signal noise σ_{z_b} is instead disciplined by the Board-learning and wage-informativeness moments, most directly v_1 (the tenure gradient in the dispersion of wage fixed effects).

4.2.3. Model validation

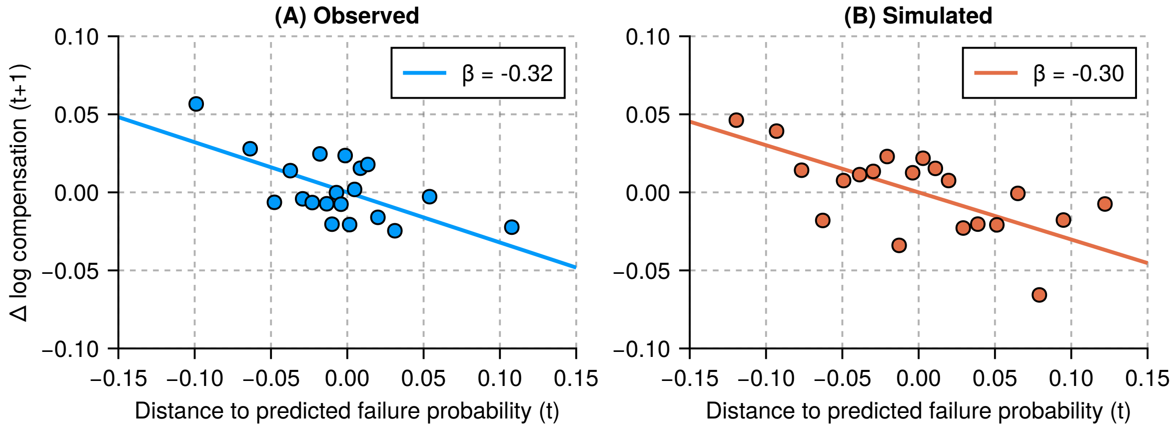
Confidence in the model's implications will be strengthened if the simulated and observed data produce similar results when examining features of the data not targeted during estimation.

Dynamics of CEO pay around Say-on-Pay failures. Figure 2 displays dynamics of CEO pay around SOP failure using a difference-in-differences specification (with SOP failure as the "treatment"). The model matches the empirical pattern of an increase in pay the year of failure followed by a decline the following year. In the model, SOP failure is incurred by a revision downwards in Shareholder beliefs about CEO type; pay thus stays below pre-failure levels for up to four years of tenure post failure.

Changes in CEO pay Following Say-on-Pay disapproval. Figure 3 displays how the observed and simulated data compare in terms of changes in CEO compensation (from t to $t+1$) in response to different SOP disapproval rates (in t). The slope of this relation is very similar in the model and data (though understandably more precise in the model). A key prediction of the model is that SOP disapproval conveys that shareholders believe the CEO is low-type. The figure shows that the Board endogenously internalizes shareholder beliefs into future pay decisions, and that the relation between the revision in pay and the magnitude of the adjustment downwards (as proxied by vote realization) is closely matched in the observed and simulated data.

Figure 3. Model fit: The relation between changes in CEO pay and SOP vote outcomes

This figure displays the relation between changes in CEO pay and the SOP vote outcome in both the observed and simulated data. In both panels, I display a binned scatterplot estimated from a regression of the (residualized) change in CEO pay on SOP disapproval, expressed as (residualized) distance to the predicted vote probability, given the state at time t. The data regression includes a year fixed; both regressions include CEO-spell fixed effects and a control for the output shock from the previous period.



5. Results

This section presents the results from the model estimation. I first discuss the estimated structural parameters and their economic magnitudes. I then discuss the impact of Say-on-Pay on compensation and shareholder value.

5.1. Estimated Structural Parameters and Economic Implications

Table 3 displays the estimated parameters. To aid interpretation, all unbounded parameters are presented as a percentage of shareholder value.[21]

5.1.1. The Board and Shareholder Costs to Say-on-Pay Failure

I estimate that the Board behaves as if SOP failure imposes a utility cost of 2.703% of average shareholder value. For shareholders, this cost is 1.130% of value (i.e., of a shareholder's equity stake). It is important to emphasize that these are utility costs: SOP failure does not affect value directly, but the Board and shareholders must behave as if it does for the model to match observed outcomes. In particular, a positive χ_S rationalizes why shareholders pass some SOP votes that would fail absent this cost.

These estimates highlight a central mechanism. Nonbinding SOP functions as a disciplinary technology, a form of shareholder voice, in which shareholders can punish perceived overpayment through

[21] In Appendix IA1.7, I derive the model's benchmark measure of average shareholder value used for normalization. All qualitative and directional conclusions are unchanged under alternative normalizations (e.g., by average operating profits or average CEO compensation); for brevity, I report only the shareholder-value normalization in Table 3.

dissent, but internalize a cost from doing so. This interpretation is consistent with evidence that negative voting outcomes (or the threat of them) can influence corporate policies even when votes are nonbinding (e.g., Del Guercio et al., 2008), and with the broader view that shareholder intervention is itself costly (e.g., Gantchev, 2013). Further, Edmans et al. (2023) provide survey evidence of the cost: in interviews, institutional investors express their reluctance to fail SOP votes. SOP failure is viewed as a reputation cost via dissenting from management on a prominent policy.[22] Importantly, the model implies that SOP's effects operate largely through equilibrium anticipation: the compensation policy brought to a vote is itself endogenous, chosen to manage the ex ante threat of failure. This mechanism helps reconcile why SOP votes typically pass with high support.

5.1.2. Economic Implications of Other Parameters

Governance wedge. The estimated wedge parameters are $\lambda_0 = 0.318$ and $\lambda_1 = 0.026$. In the model, these parameters enter wage setting through λ_τ, which scales the effective marginal cost of compensation faced by the Board. Holding fixed beliefs and SOP incentives, a higher λ_τ lowers this marginal cost and shifts the optimal wage upward. These parameters imply that the Board behaves as if it maximizes a weighted average of shareholder profits and CEO pay, with an implicit weight

$$\nu_\tau = \frac{\lambda_\tau}{1 + \lambda_\tau}$$

on CEO pay. Under the estimated parameters, this implies $\nu_0 \approx 0.24$, $\nu_{10} \approx 0.29$, and $\nu_{20} \approx 0.34$. These magnitudes are comparable to bargaining power estimates in the executive compensation literature: Taylor (2013) finds that CEOs capture roughly half of the surplus from good news, while Barry et al. (2025) estimate pure managerial bargaining power of about 0.26.[23]

Output & CEO skill parameters. The volatility of CEO skill is 1.050% of shareholder value, implying that CEO skill matters (supporting Taylor (2010), who finds that CEO fixed effects matter greatly). The output shock volatility σ_y is 1.351% of shareholder value: a large portion of observed variation in output comes from randomness, rather than CEO skill. The curvature of CEO-driven revenues with respect to compensation is $\alpha = 0.459$. This reduced-form curvature is consistent with

[22] See Online Appendix A. Investors also feel that dissent may constitute a future monitoring cost, as Boards will engage with shareholders repeatedly in the future about changing compensation policy.

[23] In Barry et al. (2025), a manager's share of surplus rises over tenure due to labor market competition. The present model abstracts from explicit competition and outside offers, so λ_τ should be interpreted as a reduced-form governance wedge rather than pure agency; in particular, some degree of "capture" could be privately optimal for shareholders as part of an efficient retention or incentive arrangement for high-skill CEOs; these considerations are outside of the model in this paper.

span-of-control views in which managerial services scale output concavely, or incentive models in which higher pay induces greater managerial input but with diminishing marginal output gains.[24]

Private information. The standard deviations of the Board's and shareholders' private signals are 2.558% and 1.169% of shareholder value, respectively. While the Board's signal innovation is more volatile than the shareholders', the Board observes its signal prior to choosing compensation; therefore, at the strategy stage the Board's posterior beliefs are more precise. Figure A.1 plots the evolution of posterior belief volatility over tenure and shows that learning is somewhat gradual: substantial uncertainty remains through the median CEO tenure length in the sample ($\tau = 7$, Table 1).

5.2. How Much Does Say-on-Pay Impact Compensation and Value?

The estimated Board cost to SOP failure implies that SOP can discipline compensation policy, but its magnitude is not directly interpretable in terms of pay and value. To quantify the impact of SOP, I simulate a counterfactual economy in which the Board places no weight on SOP failure by setting $\chi_B = 0$ and holding all other parameters at estimated values.

Table 4 reports the resulting changes in compensation and value. On average, removing SOP increases total pay by 7.76%, implying that SOP reduces compensation by this amount. This magnitude closely aligns with Correa and Lel (2016), who find that total CEO pay falls by roughly 7% upon SOP adoption in a cross-country setting. There is meaningful heterogeneity in the pay response. SOP has a larger effect for below-average ability CEOs (10.05% versus 7.33%), for CEOs early in tenure (9.16% versus 3.86%), and in observations that fail SOP in the baseline economy (12.71% versus 7.73% for pass). The pay effect is also larger when shareholder disagreement is high (11.48% versus 7.38%), consistent with discipline being strongest when the threat of dissent is most salient.

The model implies that SOP increases shareholder value by 2.97% on average, with heterogeneity that mirrors where discipline is most consequential: value gains are largest in baseline-fail observations (4.20%) and are sizable later in tenure (3.81%). Cuñat et al. (2016) estimate that adopting SOP increases market value by roughly 5% using close adoption votes prior to Dodd–Frank; since their estimate is local to firms at the adoption margin, it is reassuring that the model's average value effect is smaller.

The contribution of this counterfactual relative to the empirical adoption evidence (Cuñat et al., 2016; Correa and Lel, 2016) is the mechanism. In the model, SOP's effect is not generated by a fixed compensation policy evaluated by shareholders; rather, the compensation policy brought to a vote is

[24]For span-of-control foundations in which managerial talent/services scale firm output concavely, see Lucas (1978) and Rosen (1982). For executive-compensation models linking pay to managerial actions and firm value (assignment and moral-hazard incentives), see the survey Edmans and Gabaix (2016).

itself endogenous. Anticipating shareholders' voting rule and the possibility of failure, the Board adjusts compensation to manage ex ante failure risk. In other words, the proposal put forth is endogenous to the equilibrium stipulated by the voting environment. This implies that high observed pass rates are consistent with discipline: SOP operates primarily through the threat of failure and the induced selection of policies likely to pass, not through frequent realized failures. Observed failure realizations are therefore an incomplete statistic for SOP's impact; the relevant object is the ex ante failure risk that the Board internalizes when setting compensation.

6. Counterfactual Analysis

In its current format, SOP in the US is an ex post advisory vote. Although it serves as both a form of intervention and of communication, both roles operate indirectly. Intervention occurs indirectly through how potential dissent disciplines the Board's decision ex ante, while vote-revealed information influences the Board's pay decisions going forward (i.e., communication does not directly impact current-period wage-setting). While the results in Section 5.2 show that SOP is already effective, it is not clear that its prevailing design strikes the shareholder-optimal balance between communication and intervention. In this section, I consider counterfactual voting mechanisms that alter institutional design of SOP to allow the intervention and communication aspects of dissent to directly impact the Board's decision-making within the same period.

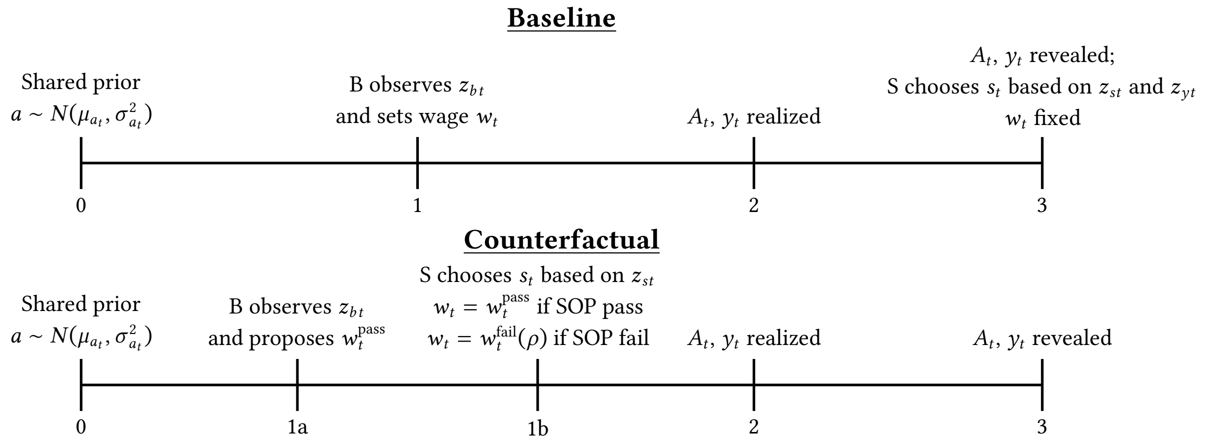
These counterfactuals are guided by a simple view of what would happen if a failed vote directly impacted the realized compensation package. An interventionist design tilts towards a more disciplinary technology: a "no" vote restricts the firm's compensation choice to a pre-defined level and thereby creates a credible anchor that may shape incentives ex ante. A more communicative design instead treats rejection as an information event: dissent reveals shareholder views about the CEO and pay package, and the Board is induced to incorporate the revealed information within the same wage-setting period.

Rather than impose that the shareholder-optimal institution must lie at either extreme, I allow for the possibility that effective governance may require some commitment and some responsiveness. This premise aligns with the broader authority and delegation literature because the ex post rule shapes ex ante incentives—affecting how the Board sets pay and how strongly it caters to shareholders' preferences and beliefs in anticipation of the vote (e.g., Aghion and Tirole, 1997). Thus, an interior rule that blends some degree of discipline with information incorporation may dominate either extreme.

Formally, let w_t^{bind} denote the wage implemented after rejection under a purely interventionist or binding rule; in my counterfactuals, this corresponds to reversion to the previously approved wage

Figure 4. Counterfactual model timeline

This figure displays the within-period model timeline of the counterfactual. The top timeline ("Baseline") displays the timeline from the main version of the model, the bottom timeline ("Counterfactual") displays the counterfactual timeline.



w_{t-1} (following SOP standards used in some countries, such as Switzerland). Let w_t^{info} denote the information-optimal wage that would be implemented after rejection under a purely communicative rule; i.e., the wage that is optimal under the posterior beliefs induced by the voting outcome when the Board fully incorporates vote-revealed information within the same period.

To consider the efficacy of mixed intervention–communication designs, I introduce a reduced-form parameter ρ that captures how strongly a failed vote anchors the post-failure outcome to the binding wage versus allowing adjustment for information. I model the effective wage choice upon failure as

$$w_t^{\text{fail}}(\rho) = \rho\, w_t^{\text{bind}} + (1-\rho)\, w_t^{\text{info}}, \qquad \rho \in [0,1], \tag{31}$$

so that $\rho = 1$ delivers a fully binding reversion (maximal intervention) and $\rho = 0$ delivers full within-period information-responsive updating (maximal communication). Intermediate values $0 < \rho < 1$ correspond to partial-binding designs.

The interpretation of ρ is reduced-form: it summarizes the institutional details of post-failure implementation into a single measure of how much discretion remains after dissent. Higher ρ strengthens shareholder control in the failure state and thereby increases ex ante discipline, while lower ρ preserves discretion and increases the scope for incorporating the information conveyed by the vote. Varying ρ within the estimated model therefore allows shareholder value to be evaluated as a function of the discipline–responsiveness tradeoff.[25]

[25] As a benchmark, the UK directors' pay confirmation regime is a hybrid of intervention and communication. Shareholders cast a binding vote on the remuneration policy, but rejection does not trigger purely mechanical adjustment: the the firm reverts to the last approved policy and the Board must return with a revised proposal for renewed shareholder approval. This design both activates an anchor and creates a structured channel through which dissent shapes policy choice.

6.1. The Vote and Compensation Decisions

This section gives a conceptual outline of how these counterfactuals change Board and Shareholder decision-making. The technical description is relegated to Appendix B. As in Section 3.4, the Board and Shareholder come into period t with the shared prior about CEO ability $a \sim N\left(\mu_{at}, \sigma_{at}^2\right)$. Instead of setting a finalized compensation package, B proposes a wage level. The Shareholder votes on the Board's proposed pay. The counterfactual implemented wage takes the form

$$w_t = \begin{cases} w_t^{\text{pass}} & = \tilde{\omega}_\rho(\mu_{at}, \sigma_{at}^2, w_{t-1}) & \text{if SOP pass} \\ w_t^{\text{fail}}(\rho) & = \rho\, w_{t-1} + (1-\rho)\, w_t^{\text{info}} & \text{if SOP fail} \end{cases}, \qquad \rho \in [0,1]. \tag{32}$$

When the vote passes, the CEO receives the Board's proposed wage, determined by policy function $\tilde{\omega}_\rho(\cdot)$ and set according to Board beliefs after the Board receives its signal z_{bt}, as in the baseline.

Figure 4 displays how the counterfactual changes the structure of the vote. Relative to the baseline, a failed vote has a direct effect on implemented compensation policy. After receiving the signal z_{bt}, the Board proposes w_t^{pass}. The outcome of the vote determines whether the CEO receives the passing or failing wage. Because failure affects within-period implementation, shareholders choose vote sensitivity s_t based on the pass–fail payoff difference, and the Board selects w_t^{pass} anticipating this response. Varying ρ changes the equilibrium wage and voting policies and, therefore, shareholder value.

Discipline vs. information. The counterfactual allows for a decomposition of effects into discipline and information channels. Holding ρ fixed, I compute (i) the total effect of moving from advisory SOP to the counterfactual regime, (ii) a discipline-only effect that keeps failure consequential but prevents within-period learning from dissent, and (iii) an information effect equal to the incremental change from reintroducing learning. Intuitively, the discipline-only effect is a counterfactual which forces the Board to act “as if" the convex combination wage in (32) is the binding wage. When $\rho = 0$ ($\rho = 1$), there is no disciplinary (informational) effect. Appendix B describes the decomposition in full.

6.2. Results

Table 5 reports the results of the counterfactuals. For each $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$, I re-solve the model and simulate outcomes using the estimated structural parameters, altering only the post-failure implementation rule implied by ρ. Each column therefore corresponds to a distinct counterfactual vote design. For each counterfactual, I report the implied SOP failure rate and the average changes in CEO pay and shareholder value relative to the baseline model. I also decompose the pay and value effects into the contributions attributable to the discipline and information channels.

Under the information-only design ($\rho = 0$), a failed vote triggers the wage that is optimal under the posterior induced by the vote. Relative to the prevailing advisory SOP, this delivers only a modest increase in value in Table 5 (column 1, about +0.36%) because it primarily improves outcomes in the relatively small set of states in which the vote fails, otherwise the newly revealed information is not incorporated within-period. Put differently, this regime strengthens the advisory role of the vote, but it leaves the board's ex ante incentives largely unchanged.

The fully binding rule ($\rho = 1$) makes rejection maximally interventionist: failure implements the previous wage rather than allowing it to adjust information revealed since the previous period. This design reduces value sharply (column 5, about −5.40%), despite lowering pay on average, because it induces informationally-inert compensation in states where within-period updating is most valuable. Moreover, because the prospect of this wage rigidity reduces the incentive to dissent (particularly in periods when failure would be most likely in the baseline), this reshapes proposed compensation even along the high-support path.

These results are consistent with Levit and Malenko (2011): nonbinding voting is a communication device whose effectiveness depends on whether the decision maker has incentives to respond to the information embedded in the vote. In the information counterfactual above, the Board is induced to incorporate information directly into the within-period wage, but the gains for shareholders are only moderately positive because, in the baseline model, the advisory vote already transmits most of the relevant information ex ante: the Board internalizes how the vote will respond when setting the wage, so strengthening the informational content of the vote largely shifts the timing of belief updating rather than materially changing outcomes. At the opposite extreme, binding rules need not dominate when they are rigid and cannot adapt to heterogeneity or to information.

The largest gains occur for intermediate values (e.g., $\rho = 0.25$ or $\rho = 0.5$), where the post-failure rule blends an anchor with scope for within-period updating. In Table 5, these regimes increase value by roughly 3 to 3.5%, and the decomposition shows that most of the gain comes from the discipline component, with a smaller but positive contribution from information.

This pattern aligns closely with the mechanism emphasized in Levit (2020): more intervention power is not monotonically better because the power to intervene can reduce the effectiveness of communication. Interpreted through that lens, ρ acts as a lever that limits how aggressively the vote design substitutes intervention and communication: intermediate ρ is valuable because it makes the vote consequential enough to tighten incentives while remaining sufficiently adaptive that the vote's information can be incorporated. This is economically similar to the sense in which partial intervention can dominate pure communication or pure intervention in Levit (2020). Importantly, these results echo the idea that communication and intervention can interact in positive ways for shareholders.

6.3. Policy Implications

Within the estimated model, the shareholder-welfare-maximizing institutional design is neither a purely advisory vote (which adds little beyond existing ex ante discipline) nor a purely binding vote (which can be value-destroying). Instead, shareholder value is highest under an anchored “revise-and-resubmit" regime: dissent should have bite, but the post-failure outcome should remain information-responsive.

One policy recommendation is that a low-support outcome should lead to two requirements. First, it activates an anchor that constrains compensation on the failure path (e.g., continuation of the last approved policy or an interim cap on total pay growth). Second, it imposes an obligation on the Board to revise and resubmit a new package that explicitly responds to the information revealed by dissent, with the vote outcome and the firm's response disclosed to investors.

Because these counterfactuals constitute policy regime changes, the conclusions suffer from the Lucas critique, to the extent that preferences are endogenous to the voting regime itself. Accordingly, the experiments should be viewed as how partial-equilibrium changes in institutional design (i.e., holding preferences fixed) affect vote outcomes and pay, and the interaction between the two, and any policy recommendations should be interpreted under these potential shortcomings.

7. Conclusion

Say-on-Pay (SOP) is a prominent shareholder voice mechanism directed at executive compensation, yet its nonbinding nature and high support rates leave its impact as a governance mechanism ambiguous. This paper clarifies both the magnitude and mechanism of SOP's impact by estimating a structural model in which the Board and shareholders can differ in preferred pay levels and their aversion to failed SOP votes. In equilibrium, the Board sets pay anticipating how shareholders translate realized compensation and information into dissent. Although only about 6% of votes fail, the estimated model implies that SOP reduces CEO pay by 7.76% and increases shareholder value by 2.97% on average.

The key implication is that the proposal put forth is endogenous to the equilibrium stipulated by the voting environment. This implies that high observed pass rates are consistent with discipline: shareholder voting can operate primarily through the threat of failure and the induced selection of proposals likely to pass, rather than through frequent realized failures. This logic applies broadly to other corporate policies subject to shareholder approval, such as equity issuance or major transactions.

I then use the estimated model to evaluate alternative vote designs that vary how dissent affects wage-setting, from a regime in which the vote serves primarily as communication to one in which they serve primarily as intervention. A fully binding vote can be value-destroying because it is rigid: mechanically tying pay to the last approved policy limits the incorporation of new information about CEO ability and alters equilibrium incentives by reducing shareholders' willingness to dissent. By contrast, hybrid designs that combine a credible post-failure anchor while allowing for information may best benefit shareholders. More generally, the results suggest that effective shareholder voting should combine intervention and communication by making dissent consequential without locking the firm into an informationally stale outcome.

Future research could integrate a strategic voting game among heterogeneous shareholders with policy-setting by the corporate decision-maker, making both the probability of failure and the informational content of the vote equilibrium objects. This would help quantify how, for example, coordination among shareholders or pivotality effects amplify or mute the vote's disciplinary effect.

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Table 1. Summary statistics

This table reports descriptive statistics for the variables used in the model estimation. The sample merges Compustat, Execucomp, and ISS for the fiscal years 2011–2024. All dollar variables are deflated to 2018 dollars. "Vote against SOP" is the fraction of votes cast against the Say-on-Pay proposal. "SOP failure" and the other indented rows are indicators equal to one when the vote-against share exceeds 30%, 20%, and 50%, respectively. "Total compensation" is total expected pay (in millions, i.e. TDC1). "Current compensation share" is current compensation divided by total compensation. "Change in total pay" is the change in (log) total compensation from the prior fiscal year, within CEO spell. "CEO tenure" is years since the CEO assumed office, and "Length of CEO tenure" is the total duration of the CEO's spell at the firm (one observation per CEO–firm spell). "Assets", "Revenues", and "Market capitalization" are in billions; market capitalization is measured at the end of the fiscal year. "Return on assets" is operating income divided by assets, and "Profit margin" is operating income divided by sales.

Variable	Unit	N	Mean	Std dev	25%	50%	75%
Say-on-Pay							
Vote against SOP	%	13,109	0.082	0.099	0.025	0.046	0.085
SOP failure: $\mathbb{1}$[More than 30% against]	{0,1}	13,109	0.061				
$\mathbb{1}$[More than 20% against]	{0,1}	13,109	0.105				
$\mathbb{1}$[More than 50% against]	{0,1}	13,109	0.018				
CEO							
Total compensation (TDC1)	$mil	13,109	7.469	7.815	3.041	5.643	9.800
Current compensation share	%	13,109	0.228	0.188	0.111	0.166	0.269
Change in total pay	%	11,659	0.019	0.420	-0.101	0.016	0.177
CEO tenure	years	13,109	6.278	5.733	2	5	9
Length of CEO tenure	years	2,324	8.725	6.159	4	7	12
Firm							
Assets	$bil	13,109	31.392	162.531	1.281	4.084	13.643
Revenues	$bil	13,109	10.819	33.231	0.824	2.323	7.628
Market capitalization	$bil	13,109	18.896	74.511	1.251	3.454	12.044
Return on assets	%	13,109	0.124	0.090	0.068	0.117	0.170
Profit margin	%	13,109	0.208	0.161	0.103	0.175	0.289

Table 2. Model fit

This table displays the estimated model's fit. Following the identification strategy in Section 4.1, I estimate the model from Section 3 using the simulated method of moments, which is described in detail in Appendix IA2. I display data and model moments (under the estimated optimal parameter vector), and a t-test of the difference between the observed and simulated data moments.

	Description	Notation	Data	Model	t-stat
Output & CEO skill					
(1)	Average log output	$\mathbb{E}[\log y]$	7.859	7.894	−0.636
			(0.038)	(0.041)	
(2)	Elasticity of output to wage	y_1	0.284	0.158	5.364
			(0.017)	(0.016)	
(3)	Output FE variance	$\text{Var}(\mu^y)$	0.495	0.528	−1.131
			(0.020)	(0.021)	
(4)	Output residual variance	$\text{Var}(\epsilon^y)$	0.267	0.498	−5.953
			(0.028)	(0.027)	
(5)	Cost share	k_1	0.366	0.380	−0.023
			(0.586)	(0.022)	
Shareholder					
(6)	SOP failure rate	$\mathbb{E}[\text{SOP fail}]$	0.061	0.059	0.461
			(0.003)	(0.002)	
(7)	SOP fail wage sensitivity	s_1	0.122	0.099	2.707
			(0.006)	(0.006)	
(8)	SOP fail output shock sensitivity	s_2	−0.043	−0.073	4.162
			(0.006)	(0.004)	
(9)	SOP fail output FE sensitivity	s_3	−0.050	−0.109	5.861
			(0.008)	(0.006)	
(10)	Excess vote-against variance	$\text{Var}\left(\tilde{V}^{\text{against}}\right)$	0.116	0.121	−0.643
			(0.006)	(0.004)	
Board					
(11)	Average log wage	$\mathbb{E}[\log w]$	1.659	1.696	−1.374
			(0.019)	(0.020)	
(12)	Change in log wage (SOP fail)	b_1	0.334	0.386	−1.392
			(0.029)	(0.023)	
(13)	Wage growth over tenure	b_2	0.038	0.033	1.406
			(0.002)	(0.002)	
(14)	Wage FE variance	$\text{Var}\left(\mu^b\right)$	0.571	0.553	0.745
			(0.016)	(0.017)	
(15)	Log wage autocovariance	$\text{Cov}(\log w, \log w_{-})$	0.594	0.650	−1.938
			(0.021)	(0.020)	
(16)	Wage FE variance tenure slope	v_1	0.008	0.007	0.253
			(0.002)	(0.002)	

Table 3. Estimated parameters

This table displays estimates of the parameters that drive the model in Section 3, given the estimation strategy in Section 4. Unscaled parameters are displayed as a percentage of average shareholder value (i.e., % s.v.), with standard errors derived via the delta method, see Appendix IA1.7 for details. The value of each parameter is displayed, along with its standard errors in parentheses.

Description	Unit	Notation	Value	
Output & CEO skill				
Prior std dev of CEO ability	% s.v.	σ_0	1.050%	(0.053%)
Std dev of productivity shock	% s.v.	σ_y	1.351%	(0.023%)
Output—CEO wage elasticity	$(0, 1)$	α	0.459	(0.026)
Scaling factor (output)	log $mil	$\log \eta$	7.124	(0.058)
Scaling factor (cost)	$(0, 1]$	κ	0.380	(0.022)
Shareholder				
Std dev of Shareholder signal	% s.v.	σ_{z_s}	1.169%	(0.088%)
Shareholder SOP failure cost	% s.v.	χ_S	1.130%	(0.206%)
Board				
Std dev of Board signal	% s.v.	σ_{z_b}	2.558%	(0.101%)
Board SOP failure cost	% s.v.	χ_B	2.703%	(0.126%)
CEO board capture (constant)	$[0, 1)$	λ_0	0.318	(0.042)
CEO board capture (growth)	$(-1, 1)$	λ_1	0.026	(0.002)
CEO wage adjustment cost	% s.v.	c_w	1.157%	(0.175%)

Table 4. The impact of Say-on-Pay on compensation and shareholder value

This table analyzes the impact of SOP on compensation policy and shareholder value. I simulate a counterfactual (using the same sequence of shocks as in the baseline estimation) in which the Board cost to SOP failure (χ_B) is set to zero while holding other parameters constant, effectively analyzing compensation policy as if SOP did not exist. The first row displays the average counterfactual percentage change in the wage level; the second displays the change in shareholder value from (22). The first column displays the average percentage changes for the entire simulated sample. The remaining columns split the sample based on cross-sectional characteristics. In columns two and three, “Low Ability" means that the CEO’s type a is less than zero (i.e., below the average); “High" ability means that the CEO’s type a is greater than zero. In columns four and five, “Early" refers to first five years of tenure (“Late" refers to years of tenure beyond five). In columns six and seven, “Fail" refers to observations in which the SOP vote would fail in the baseline; “Pass" refers to the opposite. In columns eight and nine, “Disagree" is defined as the difference between the Board and Shareholder beliefs at the time that they set their strategies: $\mu_{bt|z_b} - \mu_{at}$; “Low" refers to observations when Disagree is in the bottom 3 quartiles and “High" refers to observations in the top quartile.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
		Ability		**Tenure**		**SOP vote**		**Disagree**	
	Full sample	Low	High	Early	Late	Pass	Fail	Low	High
Total pay	+7.76%	+10.05%	+7.33%	+9.16%	+3.86%	+7.73%	+12.71%	+7.38%	+11.48%
Firm value	-2.97%	-3.25%	-2.94%	-2.67%	-3.81%	-2.95%	-4.20%	-2.98%	-2.83%

Table 5. Counterfactual voting mechanisms

This table reports a counterfactual experiment that varies the response of compensation to a failed Say-on-Pay (SOP) vote along the discipline–information trade-off, as described in Section 6 and Appendix B. In the counterfactual model, the Board proposes compensation and shareholders vote before pay is implemented. If the vote passes, the CEO receives the proposed wage. If the vote fails, the implemented wage follows a partial-binding response rule

$$w_t = \rho\, w_{t-1} + (1-\rho)\, w_t^{\text{info}},$$

where w_t^{info} is the within-period information-optimal wage under the posterior that incorporates the shareholder signal revealed by vote failure. The table reports results across regimes indexed by ρ: $\rho = 0$ corresponds to full information incorporation ($w_t = w_t^{\text{info}}$ upon failure), while $\rho = 1$ corresponds to a purely binding design ($w_t = w_{t-1}$ upon failure). Intermediate values $\rho \in (0, 1)$ represent partial binding. The first row reports the counterfactual SOP failure rate. The remaining rows report percentage changes in total CEO pay and shareholder value relative to the baseline SOP regime, computed at the observation level and then aggregated within each column. In addition, I decompose each outcome into a discipline component and an information component, as detailed in Appendix B; by construction, the discipline component is zero at $\rho = 0$, and the information component is zero at $\rho = 1$. To compute the counterfactuals, I re-solve the model under each value of ρ using the same estimated parameters and apply the same sequence of shocks across regimes. Because a binding or partial-binding wage is only relevant once there is a previously approved wage, I drop $\tau = 0$ observations from the analysis.

	(1)	(2)	(3)	(4)	(5)
Regime:	**Communication**		**Partial-binding**		**Binding**
	$\rho = 0$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1$
SOP fail rate	3.63%	10.13%	12.91%	13.75%	5.36%
% change in total pay					
Total effect	-0.27%	-3.96%	-3.65%	-1.04%	-1.46%
Discipline channel	+0.00%	-3.15%	-3.28%	-1.74%	-1.46%
Information channel	-0.27%	-0.80%	-0.37%	+0.70%	+0.00%
% change in shareholder value					
Total effect	+0.36%	+3.42%	+3.25%	-1.06%	-5.40%
Discipline channel	+0.00%	+3.27%	+3.15%	-1.06%	-5.40%
Information channel	+0.36%	+0.15%	+0.09%	-0.00%	+0.00%

A. Appendix Figures and Tables

Figure A.1. Board and Shareholder volatility of beliefs over CEO tenure

This figure displays the volatility Board and shareholder beliefs about CEO ability as a function of CEO tenure τ at the time at which each agent plays their strategy, using the parameter estimates from Table 3. Each volatility of beliefs is at the time that each party plays their strategy: the Board's belief volatility is $\sigma_{bt|z_b}$, as in (19), and the Shareholder's belief volatility is σ_{at} (i.e., the prior).

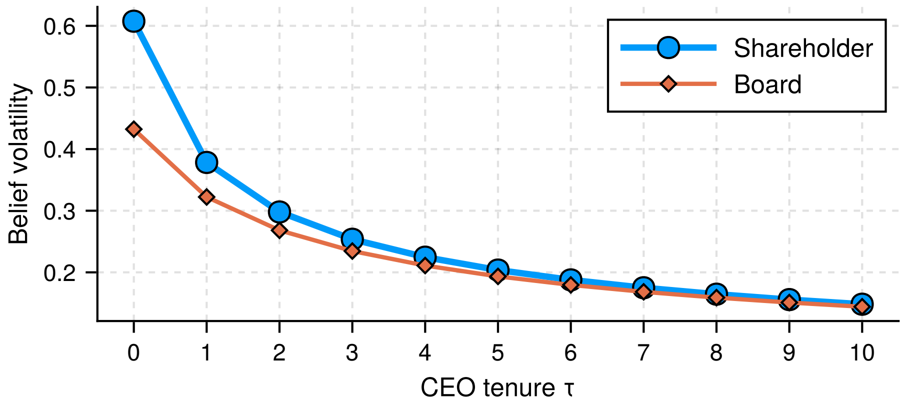


Table A.1. Moment targeting exercise

This table provides detail on targeted empirical moments. Panel A describes each moment, and lists the parameter(s) it targets. Panel B provides pairwise correlations. The correlations in Panel B are constructed via the empirical variance-covariance matrix, estimated using the influence function approach detailed in Appendix IA2.3.

Panel A. Targeted empirical moments

	Description	Notation	Target
	Output & CEO skill		
(1)	Average log output	$\mathbb{E}[\log y]$	$\log \eta$
(2)	Elasticity of output to wage	y_1	α
(3)	Output FE variance	$\text{Var}(\mu^y)$	σ_0
(4)	Output residual variance	$\text{Var}(\epsilon^y)$	σ_y
(5)	Cost share	k_1	κ
	Shareholder		
(6)	SOP failure rate	$\mathbb{E}[\text{SOP fail}]$	χ_S
(7)	SOP fail wage sensitivity	s_1	$\chi_S,\ \chi_B$
(8)	SOP fail output shock sensitivity	s_2	σ_{z_s}
(9)	SOP fail output FE sensitivity	s_3	$\sigma_{z_s},\ \chi_S$
(10)	Excess vote-against variance	$\text{Var}\left(\tilde{V}^{\text{against}}\right)$	σ_{z_s}
	Board		
(11)	Average log wage	$\mathbb{E}[\log w]$	λ_0
(12)	Change in log wage (SOP fail)	b_1	χ_B
(13)	Wage growth over tenure	b_2	λ_1
(14)	Wage FE variance	$\text{Var}\left(\mu^b\right)$	$\sigma_{z_b},\ c_w$
(15)	Log wage autocovariance	$\text{Cov}(\log w, \log w_{-})$	c_w
(16)	Wage FE variance tenure slope	v_1	σ_{z_b}

Panel B. Pairwise correlations between empirical moments

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
$\mathbb{E}[\log y]$	(1)																
k_1	(2)	-0.08															
y_1	(3)	-0.09	0.24														
$\mathrm{Var}(\mu^y)$	(4)	0.22	0.11	0.01													
$\mathrm{Var}(\epsilon^y)$	(5)	-0.15	0.29	0.19	-0.02												
$\mathbb{E}[\text{SOP fail}]$	(6)	0.00	-0.09	-0.10	0.06	0.09											
s_1	(7)	-0.05	-0.08	0.06	-0.11	0.13	0.42										
s_2	(8)	-0.05	-0.06	0.10	-0.12	0.19	0.02	0.50									
s_3	(9)	0.10	0.12	-0.05	0.19	-0.14	-0.26	-0.73	-0.76								
$\mathbb{E}[\log w]$	(10)	0.76	0.03	0.01	0.10	0.01	0.17	0.13	0.01	-0.04							
b_1	(11)	-0.08	-0.05	-0.17	-0.07	0.05	0.05	0.40	0.13	-0.25	-0.03						
b_2	(12)	-0.13	0.11	0.46	-0.01	0.11	-0.07	0.02	0.04	-0.01	-0.08	0.06					
$\mathrm{Var}(\mu^b)$	(13)	-0.06	-0.06	-0.03	0.24	-0.05	0.00	-0.09	-0.03	0.03	-0.28	-0.03	-0.05				
$\mathrm{Cov}(\log w, \log w_{-})$	(14)	-0.01	-0.02	0.04	0.23	-0.02	0.03	-0.09	-0.04	0.09	-0.23	-0.04	0.02	0.77			
$\mathrm{Var}(\tilde{V}^{\text{against}})$	(15)	-0.06	0.08	0.10	-0.12	-0.03	-0.76	-0.29	0.00	0.18	-0.13	0.01	0.11	-0.07	-0.08		
v_1	(16)	0.12	-0.04	-0.08	0.21	-0.05	0.01	-0.06	0.01	0.04	-0.03	-0.01	-0.12	0.68	0.72	-0.06	

Table A.2. Parameter–moment sensitivity for key parameters

This table presents the Andrews et al. (2017) standardized sensitivity of parameters to moments for the estimation results in Tables 2 and 3. Standardized sensitivity ranges from -1 to +1, with larger absolute magnitudes indicating higher sensitivity. For brevity, I focus on key parameters: those relating to the Board and Shareholder decisions.

	σ_{z_s}	χ_S	σ_{z_b}	χ_B	λ_0	λ_1	c_w
$\mathbb{E}$[SOP fail]	-0.01	-0.05	-0.02	0.21	0.01	-0.01	-0.21
s_1	0.37	-0.27	-0.51	0.08	0.33	0.03	-0.11
s_2	0.58	-0.56	-0.23	0.07	0.68	0.61	-0.15
s_3	-0.31	0.31	0.16	0.04	-0.31	0.02	-0.52
$\mathbb{E}[\log w]$	-0.03	0.06	-0.06	-0.10	0.02	0.04	-0.14
b_1	-0.34	0.20	-0.49	0.77	-0.20	-0.44	0.06
b_2	0.34	-0.33	-0.17	-0.16	-0.07	0.13	0.70
$\text{Var}(\mu^b)$	0.14	-0.38	0.11	0.24	0.20	-0.09	0.13
$\text{Cov}(\log w, \log w_{-})$	0.22	0.28	0.32	-0.36	0.14	0.50	0.12
$\text{Var}(\tilde{V}^{\text{against}})$	0.32	0.02	-0.01	0.12	0.05	-0.02	-0.14
v_1	-0.01	0.04	0.26	-0.06	-0.00	0.00	0.13

B. Counterfactual Appendix

This section outlines the technical details of the counterfactual in Section 6.

Timing and information. The partial-binding counterfactual modifies the within-period timing of the vote and, as a result, the information that enters the voting decision. In the baseline advisory regime (Section 3.2), the Board sets pay before output is realized, and the Say-on-Pay vote occurs at the annual meeting after output and productivity are revealed; thus the vote can condition on the public productivity signal $z_{yt} = \ln A_t$ from (11) in addition to shareholders' private signal z_{st} from (10). This is reflected in the baseline vote signal aggregation in (15) and the induced failure probability in (17).

In the partial-binding counterfactual, the vote is moved forward to occur immediately after the compensation committee meeting and before operations take place. Consequently, when voting, shareholders do not observe output and do not observe the productivity signal z_{yt} from (11). The vote therefore conditions on the proposed wage and shareholders' private signal z_{st} (and any information inferred from the proposal), but not on within-period performance realizations. This timing change is the main informational difference between the counterfactual and the baseline.

The Shareholder's problem. As in the baseline, shareholders commit ex ante to a voting policy indexed by s_t (Section 3.3), taking expectations over within-period uncertainty and integrating over the Board's private signal z_{bt} (since z_{bt} is realized before the wage proposal). The key difference from the baseline shareholder problem in (18) is that vote failure now changes implementation within the period, so shareholders internalize a pass–fail payoff difference rather than only an expected failure cost.

Let $\tilde{F}_t^{S}(s_t \times \mathrm{w})$ denote the probability of failure induced by the threshold rule, defined analogously to (17) but with the counterfactual information set at the vote: because the vote occurs before A_t and y_t are realized, $\tilde{F}_t^{S}(\cdot)$ is induced by shareholders' private signal z_{st} (and the proposal), rather than by the combined signal (z_{st}, z_{yt}) in (15). Define the firm's operating income under shareholders' information as

$$\tilde{\pi}_{st}(\mathrm{w}) = \mathbb{E}_{st}[A_t]\mathrm{w}^{\alpha} - \kappa\, \mathrm{w},$$

where the expectation is taken conditional on shareholders' information at the voting stage (which excludes the realized productivity signal z_{yt} in (11)).

Given the Board's proposed wage w_t^{pass}, shareholders choose s_t to maximize expected operating income under the wage implemented in each vote outcome:

$$\max_{s_t} \quad \left(1 - \tilde{F}_t^{\mathrm{S}}\left(s_t \times w_t^{\text{pass}}\right)\right) \times \underbrace{\int_{z_b} f(z_b)\left[\ \tilde{\pi}_{st}\left(w_t^{\text{pass}}\right)\right]\ dz_b}_{\text{Pass: CEO receives proposed wage}} \ + $$
$$F_t^{\mathrm{S}}\left(s_t \times w_t^{\text{pass}}\right) \times \underbrace{\left(\int_{z_b} f(z_b)\left[\ \tilde{\pi}_{st}\left(w_t^{\text{fail}}(\rho)\right)\right]\ dz_b \ - \ \chi_S\right)}_{\text{Fail: CEO receives } w_t^{\text{fail}}(\rho) \text{ \& S pays cost } \chi_S}. \tag{B.1}$$

Note that the chosen wage w_t^{pass} is a function of (σ_{z_b}, s_t), which is excluded for notational convenience. Relative to (18), the term $\chi_S \times \tilde{F}_t^{\mathrm{S}}(\cdot)$ is no longer the only consequence of failure: failure also changes the implemented wage from w_t^{pass} to $w_t^{\text{fail}}(\rho)$, which changes operating income directly.

The Board's problem. In the baseline, the Board chooses the implemented wage and trades off operating income, bias, adjustment costs, and the expected utility cost of SOP failure (see (20)). Under partial-binding SOP, the Board instead proposes w_t^{pass} anticipating that the implemented wage equals w_t^{pass} if the vote passes and equals $w_t^{\text{fail}}(\rho)$ if the vote fails. Because failure changes implementation, the Board's objective must account for pay and continuation values in both branches.

Formally, under partial-binding SOP the Board proposes w_t^{pass} according to

$$\tilde{V}(\mu_{at}, \tau_t, w_{t-1}) = \max_{w_t^{\text{pass}}}\left(1 - \tilde{F}_t^{\mathrm{S}}\left(s_t \times w_t^{\text{pass}}\right)\right)\left[\tilde{\pi}^{\mathrm{B}}_{at|z_{bt}}(w_t^{\text{pass}}) - AC(w_t^{\text{pass}}, w_{t-1}; \tau_t) + \tilde{V}_{at|z_{bt}}(w_t^{\text{pass}})\right] +$$
$$\tilde{F}_t^{\mathrm{S}}\left(s_t \times w_t^{\text{pass}}\right)\left[\tilde{\pi}^{\mathrm{B}}_{at|z_{bt}}\left(w_t^{\text{fail}}(\rho)\right) - AC\left(w_t^{\text{fail}}(\rho), w_{t-1}; \tau_t\right) + \tilde{V}_{at|z_{bt}}\left(w_t^{\text{fail}}(\rho)\right) - \chi_B\right], \tag{B.2}$$

where

$$\tilde{\pi}^{\mathrm{B}}_{at|z_{bt}}(\mathrm{w}) = \mathbb{E}_{at|z_{bt}}[A_t]\mathrm{w}^{\alpha} - \kappa\mathrm{w} + \lambda_{\tau}\kappa\mathrm{w},$$
$$\tilde{V}_{at|z_{bt}}(\mathrm{w}) = \mathbb{E}_{at|z_{bt}}\left[(1 - f_{\tau_t})[V(\mu_{at+1}, \tau + 1, \mathrm{w})] + f_{\tau_t} V^R\right].$$

Unlike the fully binding case, compensation can change upon failure when $\rho < 1$, so an adjustment cost may arise in the failure branch; this cost vanishes when $\rho = 1$ because $w_t^{\text{fail}}(\rho) = w_{t-1}$.

Discipline vs. information decomposition. We can decompose the counterfactual impacts on pay and shareholder value into a discipline and information channel. For a given design indexed by ρ, let X denote an outcome (e.g., CEO compensation w or shareholder value V). For each simulated observation i, define: (i) X_i^0 under the baseline advisory regime, (ii) $X_i^D(\rho)$ under a discipline-only counterfactual that preserves the intervention component of the ρ-rule but shuts down within-period information updating, and (iii) $X_i^T(\rho)$ under the full counterfactual.

Counterfactual effects are measured as expected log changes relative to the baseline:

$$\Delta^T(\rho) \equiv \mathbb{E}\left[\log\left(\frac{X^T(\rho)}{X^0}\right)\right], \qquad \Delta^D(\rho) \equiv \mathbb{E}\left[\log\left(\frac{X^D(\rho)}{X^0}\right)\right], \qquad \Delta^I(\rho) \equiv \mathbb{E}\left[\log\left(\frac{X^T(\rho)}{X^D(\rho)}\right)\right]. \tag{B.3}$$

This definition yields the exact additive decomposition

$$\Delta^T(\rho) = \Delta^D(\rho) + \Delta^I(\rho), \tag{B.4}$$

so the information component can be recovered as the residual $\Delta^I(\rho) = \Delta^T(\rho) - \Delta^D(\rho)$.

Intuitively, $\Delta^D(\rho)$ simulates a “false" counterfactual in which the binding wage is a scalar that represents the convex combination of the full-information and lagged wages, but I disallow any learning to occur (i.e., the Board takes the proposed wage as the binding wage, but does not incorporate any information into their decision). $\Delta^I(\rho)$ is then the residual from comparing $X^D(\rho)$ to $X^T(\rho)$, which thus isolates the information component.

Solution. I solve the counterfactual by simulation, holding primitives fixed and using the same sequence of shocks as in the baseline (signals, productivity/output, and separations). First, I solve for the full-information wage - the wage the Board would pay under the posterior stipulated after incorporating vote-revealed information, giving w_t^{info}. For each ρ, I compute the Board’s optimal proposal w_t^{pass} from (B.2), taking into account that the vote occurs before z_{yt} is realized and that failure implements $w_t^{\text{fail}}(\rho)$; this delivers simulated wages and shareholder value under the full counterfactual, $X^T(\rho)$. To measure discipline, I rerun the same shock paths in a “discipline-only” economy that preserves the ρ (the same $w_t^{\text{fail}}(\rho)$ anchor) but eliminates any belief updating from the vote, so outcomes differ only through implementation, not information; this produces $X^D(\rho)$. I then compute $\Delta^T(\rho)$ and $\Delta^D(\rho)$ from simulated log changes and recover the information component as $\Delta^I(\rho) = \Delta^T(\rho) - \Delta^D(\rho)$.

Internet Appendix

IA1. Model Appendix

IA1.1. Microfoundation of Board wage-setting

This appendix provides a simple, one-period microfoundation for two reduced-form objects used in the main text: (i) a concave mapping from the level of expected pay to the CEO's productive input, and (ii) a force that induces pay above the shareholder-optimal benchmark. This provides one interpretative microfoundation for the production technology and difference in preferences in the main text.

CEO input choice and the pay–output reduced form. The Board offers a linear contract that pays ω per unit of CEO input n, so total pay is $w \equiv \omega n$. The CEO chooses n to maximize

$$u(\omega, n) = \omega n - n^{1/\gamma}, \qquad \gamma \in (0, 1).$$

The CEO's best response solves $\omega = \frac{1}{\gamma} n^{1/\gamma - 1}$. Substituting into $w = \omega n(\omega)$, eliminating ω yields and absorbing constants into the output scale allows us to write the productive input as a function of the total wage level:

$$n(w) = w^{\gamma}.$$

Output is produced according to

$$y(A, n) = \eta A n^{\beta}, \qquad \beta > 0,$$

so with $n(w) = w^{\gamma}$ we obtain the main-text mapping

$$y = \eta A w^{\alpha}, \qquad \alpha \equiv \beta\gamma \in (0, 1).$$

Shareholder-optimal wage. Let operating profits be

$$\pi(A, w) = A w^{\alpha} - \kappa w.$$

The shareholder-optimal wage (absent any wedge) is

$$w_S(A) = \arg\max_{w>0} \ \pi(A, w) = \left(\frac{\alpha A}{\kappa}\right)^{\frac{1}{1-\alpha}}.$$

Reference-wage or fairness constraint. Suppose the Board faces a reference-wage (or fairness/liquidity) requirement that the CEO's pay level must satisfy

$$w \geq (1 + b_{\tau})\,\overline{w}(A), \qquad b_{\tau} \geq 0,$$

where b_{τ} may vary with CEO tenure. If the constraint is slack, the Board implements $w_S(A)$. If it binds, the implemented wage is

$$w_B(A) = (1 + b_{\tau})\overline{w}(A).$$

A natural benchmark is $\overline{w}(A) = w_S(A)$, so that binding implies

$$w_B(A) = (1 + b_{\tau}) w_S(A) > w_S(A).$$

Because $n(w)$ and $y(A, w)$ are increasing in w, binding reference-wage requirements imply

$$w_B > w_S \Rightarrow n_B > n_S \Rightarrow y_B > y_S.$$

But since w_S maximizes $\pi(A, w)$, any $w_B \neq w_S$ lowers shareholder profits; in particular, $w_B > w_S$ implies $\pi(A, w_B) < \pi(A, w_S)$. Thus, the reference-wage constraint generates greater CEO input (and output) relative to the shareholder-optimal benchmark, as in the main text.

Discussion. In the static version of the main model, a Board–shareholder preference wedge λ_τ that lowers the Board's effective marginal cost of pay delivers a wage of the form

$$w(\lambda_\tau) = \left(\frac{\alpha A}{\kappa(1-\lambda_\tau)}\right)^{\frac{1}{1-\alpha}} = (1-\lambda_\tau)^{-\frac{1}{1-\alpha}}\, w_S(A).$$

Hence a binding reference-wage distortion is observationally equivalent to a governance wedge under

$$1 + b_\tau = (1-\lambda_\tau)^{-\frac{1}{1-\alpha}} \qquad \Longleftrightarrow \qquad \lambda_\tau = 1 - (1+b_\tau)^{-(1-\alpha)}.$$

This provides an interpretation for λ_τ as capturing (among other forces) level-based reference-wage/fairness constraints that push compensation above the shareholder-optimal level.

This interpretation also clarifies identification: under $y = \eta A^\alpha$, the log–log elasticity of revenues with respect to pay equals α, so the auxiliary revenue–pay slope used in estimation directly disciplines the curvature parameter governing how pay levels translate into productive input in the model.

IA1.2. Representative shareholder with correlated information

This appendix provides a simple informational justification for representing a diffuse shareholder base by a single "representative" shareholder. The key assumptions are that shareholder information has a common component (e.g., through proxy advice and shared information sources), so private signals are correlated across investors, and that shareholders are atomistic.

Proposition IA.1 (Informational equivalence)**.** *Consider a continuum of atomistic shareholders $i \in [0,1]$ who each observe a signal*

$$z_i = \overline{z} + \varepsilon_i, \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma_\varepsilon^2),$$

where the common component satisfies

$$\overline{z} = a + \nu, \qquad \nu \sim N(0, \sigma_{\overline{z}}^2).$$

Suppose shareholders use a symmetric cutoff rule: vote against *iff $z_i \le k(w)$, where $k(w)$ is an increasing function of the proposed wage (e.g., $k(w) = s \times w$). Let p denote the fraction of votes cast against. Then, as the shareholder base becomes large,*

$$p = \Phi\left(\frac{k(w) - \overline{z}}{\sigma_\varepsilon}\right) \quad a.s.$$

Hence the vote share p is a sufficient statistic for the common signal $\overline{z}$; equivalently, observing p is informationally equivalent to observing the aggregate signal

$$z_S \equiv \overline{z} = k(w) - \sigma_\varepsilon\, \Phi^{-1}(p),$$

with $z_S = a + \nu$ and $\nu \sim N(0, \sigma_{\overline{z}}^2)$. Therefore the shareholder base can be modeled as a representative shareholder who observes z_S and applies the same cutoff rule.

Proof. Conditional on $\overline{z}$, each shareholder votes against with probability $q(\overline{z}) = \Pr(z_i \le k(w) \mid \overline{z}) = \Phi((k(w) - \overline{z})/\sigma_\varepsilon)$. With atomistic ownership, the realized vote share converges to this probability by a law-of-large-numbers argument. Intuitively, the probability of observing a vote result not stipulated by the aggregate signal vanishes in the limit (i.e., the purely idiosyncratic errors do not play a role). Since $q(\cdot)$ is strictly monotone in $\overline{z}$, p is equivalent to $\overline{z}$ via inversion. See Barry et al. (2026) for a more detailed proof. ■

Discussion. The correlated information structure can be interpreted as proxy-advisor recommendations shifting the common component $\overline{z}$ (and/or reducing $\sigma_{\overline{z}}^2$ by making shareholder information more correlated).

An important point is that this microfoundation is purely informational: it does not model strategic interaction among heterogeneous shareholders. Rather, it provides a simple way to take realistic features of the shareholder base, most importantly, correlated information arising from common research, stewardship norms, and proxy-advisor input, and shows that, under atomistic ownership and a symmetric cutoff rule, the realized vote share is of a single common signal. Thus, the dispersed shareholder base can be summarized by a "representative" shareholder who observes an aggregate signal and applies the same threshold strategy.

In the main text, I therefore treat shareholders as a representative voter not because investors are literally representative, but because the equilibrium voting outcome can be written as a reduced-form failure probability ($\Pr(\text{fail} \mid w, \mathcal{I}) = F(\text{threshold}(w, \mathcal{I}))$) driven by this aggregate signal. This keeps the model focused on the paper's main mechanism—how boards internalize the threat of dissent when setting pay.

IA1.3. Microfoundation of commitment to costly failed votes

This subsection provides a short relational-contract interpretation (Bull, 1987) for the model's assumption that shareholders can commit to a voting rule even though a realized failed vote is costly. Consider an infinitely repeated interaction between the Board (B) and shareholders (S). In each period, B chooses either a disciplined wage w_P or an overpaying wage $w_F > w_P$. If S fails the vote, S pays a one-time cost X_S and B pays X_B, and play continues. If S does not fail after observing w_F, we assume the disciplinary threat loses credibility and B pays w_F in all future periods.

A simple grim-trigger strategy supports discipline: S fails whenever $w = w_F$, and otherwise does not fail. Failing is optimal for S if the current cost is no larger than the discounted value of preserving the low-wage continuation:

$$X_S \leq \delta\big(E[V_D] - E[V_{ND}]\big) = \frac{\delta}{1-\delta} E[w_F - w_P].$$

Thus, costly vote failure is credible when the present value of future discipline exceeds the contemporaneous cost.

Discussion. The inequality

$$X_S \leq \frac{\delta}{1-\delta} E[w_F - w_P]$$

formalizes that even if a realized failed vote is costly ex post, failure can be self-enforcing because it preserves the credibility of future discipline and thereby sustains a lower-wage continuation path. In the main text, I capture this logic in reduced form by assuming shareholders commit ex ante to a voting policy that pins down an anticipated probability of failure, which the Board internalizes when setting pay. Absent such commitment, shareholders would have an incentive to threaten failure to influence compensation ex ante and then renege at the meeting to avoid the cost, eliminating SOP's disciplining power. Institutionally, repeated interaction and publicly articulated stewardship and voting guidelines provide a natural channel through which such credibility can arise, even when individual votes are advisory and costly.

IA1.4. Evolution of Board and Shareholder Beliefs

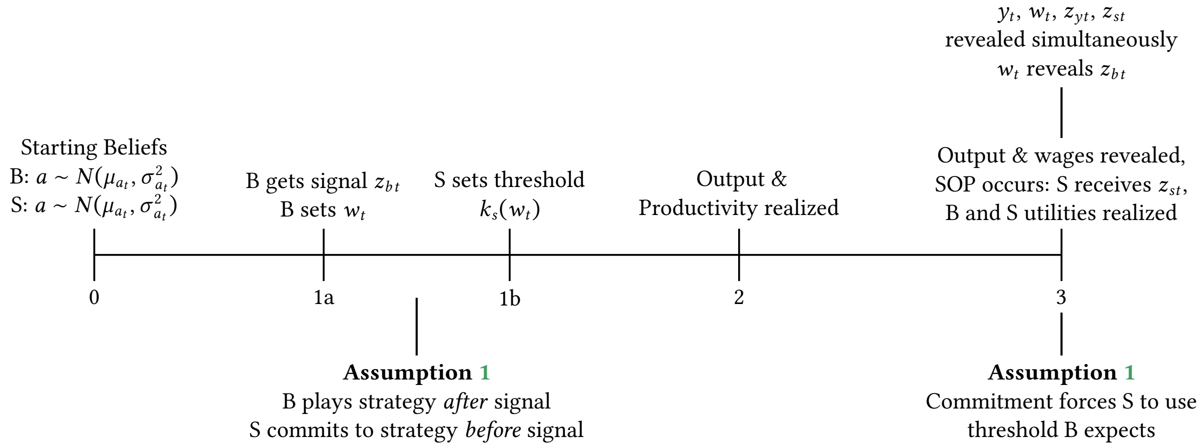
I first detail two Propositions, which define how beliefs update in the model. Then I define exactly how Board and shareholder beliefs change within each period.

IA1.4.1. Evolution of Beliefs Period to Period

Prop. IA.2 shows how beliefs change from t to $t + 1$. Prop. IA.3 describes the distribution of next period beliefs given today's beliefs, which is used when the Board calculates their (expected) continu-

Figure IA.1. Model timeline with exact time that strategies are played

This figure displays a more detailed model timeline, with mappings to the relevant assumptions, and when the Board and Shareholder play their strategies. See Figure 1 in the main text for the timeline as it maps to real-world outcomes. See Appendix IA1.5.



ation value. Figure IA.1 displays a detailed timeline that illustrated belief updating and the timing of strategies.

Proposition IA.2. *From period t to $t+1$, the variance of beliefs for both the Board and shareholders declines deterministically according to*

$$\sigma_a^2(\tau+1) = \left[\sigma_a^{-2}(\tau) + \sigma_{z_b}^{-2} + \sigma_{z_s}^{-2} + \sigma_y^{-2}\right]^{-1} \tag{IA.1}$$

where τ is the tenure of the CEO at year t. Equivalently, I can write the variance of beliefs about CEO ability entirely as a function of CEO tenure τ and model parameters

$$\sigma_a^2(\tau) = \sigma_0^2\left[1 + \tau\left(\kappa_{z_b}^{-1} + \kappa_{z_s}^{-1} + \kappa_y^{-1}\right)\right]^{-1} \tag{IA.2}$$

where $\kappa_{z_b} = \sigma_{z_b}^2/\sigma_0^2$, $\kappa_{z_s} = \sigma_{z_s}^2/\sigma_0^2$ and $\kappa_y = \sigma_y^2/\sigma_0^2$

Similarly, from period t to $t+1$, the mean of beliefs for both the Board and shareholders evolves according to

$$\mu_{at+1} = \sigma_a^2(\tau+1)\left[\frac{\mu_{at}}{\sigma_a^2(\tau)} + \frac{z_{bt}}{\sigma_{z_b}^2} + \frac{z_{st}}{\sigma_{z_s}^2} + \frac{z_{yt}}{\sigma_y^2}\right] \tag{IA.3}$$

Proof. The formulas are standard results in Bayesian learning (e.g., Pastor and Veronesi, 2009; Taylor, 2010).[1] The Board and shareholder reveal their signals each period. Thus, Board and shareholders share the same beliefs about the variance from period to period. ■

Proposition IA.3. *The mean and variance of the mean of $t+1$ CEO beliefs at t are*

$$\begin{aligned}\mathbb{E}_t[\mu_{at+1}] &= \mu_{at} \\ Var_t[\mu_{at+1}] &= \sigma_a^2(\tau) - \sigma_a^2(\tau+1)\end{aligned} \tag{IA.4}$$

That is,

$$\mu_{at+1} \mid \mu_{at}, \tau \sim N\left(\mu_{at}, \sigma_a^2(\tau) - \sigma_a^2(\tau+1)\right)$$

[1] See also the internet appendix for Taylor (2010).

Proof. Again, this is a standard result in Bayesian learning, i.e., $\text{Var}(E[a \mid I'] \mid I) = \text{Var}(a \mid I) - \text{Var}(a \mid I')$, where a is a Normal process, and I and I' are information sets before and after realizations of normal signals about a. ■

IA1.4.2. Differences in Board and Shareholder Beliefs Within Period

This sections how Board and shareholder beliefs evolve within each period. As the wage and vote reveal signals z_b and z_s, B and S share the same beliefs at the beginning of t. By Prop. IA.2, I have that $\sigma^2_{bt} = \sigma^2_{st} = \sigma^2_a(\tau_t)$ from (IA.2). Let μ_{at} be the beliefs about the mean at the beginning of period t. The evolution of B and S beliefs within t is:

1. **Board beliefs after the compensation committee meeting**
   At the meeting, the Board receives signal z_{bt}, and Board beliefs update to

$$\mu_{bt|z_b} = \sigma^2_{bt|z_b}\left(\frac{\mu_{at}}{\sigma^2_a(\tau_t)} + \frac{z_{bt}}{\sigma^2_{z_b}}\right) \tag{IA.5}$$

$$\sigma^2_{bt|z_b} = \frac{\sigma^2_a(\tau_t)\sigma^2_{z_b}}{\sigma^2_a(\tau_t) + \sigma^2_{z_b}} = \sigma^2_0\left[1 + (\tau_t + 1)\kappa^{-1}_{z_b} + \tau_t(\kappa^{-1}_{z_s} + \kappa^{-1}_y)\right]^{-1} \tag{IA.6}$$

   The Board makes their wage decision based upon these beliefs.

2. **Shareholder beliefs when they commit to signal threshold k_{st}**
   At the time that S commits to voting strategy, B has beliefs $\left(\mu_{bt|z_b}, \sigma_{bt|z_b}\right)$ and S has $\left(\mu_{at}, \sigma^2_a(\tau_t)\right)$ (the prior). S can discern $\left(\mu_{bt|z_b}, \sigma_{bt|z_b}\right)$ for any z_{bt}, which they integrate out in the objective function (18).

3. **Board and shareholder beliefs about shareholders' *ex ante* signal distribution at the time of the SOP vote**
   Before the SOP vote, when the shareholders commit to their threshold, both B and S know that the shareholders' aggregated signal will be

$$Z_{syt} = a + p\varepsilon_{st} + (1-p)\varepsilon_{yt} \tag{IA.7}$$

   with $Z_{syt} \sim N\left(\mu_{at}, \sigma^2_{at} + \frac{\sigma^2_{z_s}\sigma^2_y}{\sigma^2_{z_s}+\sigma^2_y}\right)$.[2] Notice that shareholder beliefs about a do not update to $N\left(\mu_{bt|z_b}, \sigma_{bt|z_b}\right)$. This is because the timing convention in the model states that the wage w_t (and equivalently z_{bt}), productivity z_{yt} and the signal z_{st} are all revealed concurrently (see Figure IA.1).

4. **Board and shareholder beliefs after the annual shareholder meeting**
   The 10-K and compensation committee report reveals the wage to shareholders, hence reveals z_{bt}. Shareholders vote at the annual meeting and Z_{syt} and thus z_{st} are revealed. Hence, B and S beliefs update to $\left(\mu_{at+1}, \sigma^2_a(\tau_t+1)\right)$, by Prop. IA.2.

IA1.5. Assumptions about Shareholder strategy

See Section 3.3 for discussion of the Shareholder's strategy. I model this threat as an *ex ante* probability that the SOP will fail, which is increasing in the wage, and I make the following assumptions:

Assumption 1. *Shareholders commit to their voting strategy in advance of the annual shareholder meeting.*

I assume that S commits to an *ex ante* probability of vote failure to influence the Board's wage decision, even though *ex post* failure is costly. Because of the information revelation structure of the model

[2] The variance of the signal is $Var(a + p\varepsilon_{st} + (1-p)\varepsilon_y) = \sigma^2_a + p^2\sigma^2_{st} + (1-p)^2\sigma^2_y$, where $p = \frac{\sigma^{-2}_{z_s}}{\sigma^{-2}_{z_s}+\sigma^{-2}_y}$. Expanding this expression out leads to the expression for the variance.

(Figure IA.1), S sets the probability of vote failure before seeing z_{bt}, z_{st}, or z_{yt}. The threat of vote failure does not need to be revealed to the Board before the annual shareholder meeting, however commitment forces S to play the threshold strategies that the Board expects. Unlike Kakhbod et al. (2023), there is no notion of cheap talk here. Commitment means S cannot choose an *ex ante* optimal non-zero failure probability and then renege at the shareholder meeting once the Board sets their wage.[3]

Assumption 2. *Shareholders are* myopic. *That is, the SOP vote is only influenced by today, and is not a fully dynamic problem.*

Effectively, this means that shareholders play a static game, while the Board plays a dynamic one. This assumption matches reality. There is ample evidence that voting in SOPs is influenced by short-run outcomes, such as current firm or stock performance (see Fisch et al., 2018; Novick, 2019, 2020). This makes the solution method much simpler, as I only need to solve for one value function in the numerical solution.

IA1.6. Full Derivation of Model Solution

Proposition IA.4. *The Board's problem can be written as*

$$V(\mu_a, \tau, w_-) = \max_{w(z_b,s)} \quad \exp\left(\mu_{b|z_b} + 0.5\left(\sigma^2_{b|z_b} + \sigma^2_y\right)\right) w(z_b, s)^\alpha - (1-\lambda_\tau) w(z_b, s) - \chi_B F^U_{\tilde{z}_s}(s \times w(z_b, s)) - AC(w(z_b, s), w_-; \tau) + \delta_B \left[f(\tau_t) V^R + (1 - f(\tau_t)) \mathbb{E}_{b|z_b} \left[V(\mu_a', \tau + 1, w(z_b, s)) \right] \right] \tag{IA.8}$$

where

- $\mu_{b|z_b}$ *and* $\sigma^2_{b|z_b}$ *are defined in* (IA.5) *and* (IA.6),
- $\mathbb{E}_{b|z_b}[V] = F^U_{\tilde{z}_s}(s \times w)$ *where* $F^U_{\tilde{z}_s}$ *is the CDF as in* (16)
- $AC(w, w_-; \tau)$ *(adjustment cost) is defined in* (5),
- $f(\tau_t)$ *are CEO tenure-specific hazard rates, with* $f_0 = 0$ *and* $f(\tau_t) = 1$
- $V^R = V(\mu_0, 0, 0)$ *as in* (21)
- $\mu_a' \mid \mu_{b|z_b}, \tau \sim N\left(\mu_{b|z_b}, \sigma^2_{b|_{z_b}} - \sigma^2_a(\tau+1)\right)$ *from Prop. IA.3*

The shareholder's problem can be written as

$$\max_s \int_{z_b} f(z_b \mid \mu_a, \sigma_a^2) \left[\exp\left(\mu_a + 0.5\left(\sigma^2_{b|z_b} + \sigma^2_y\right)\right) w(z_b, s)^\alpha - w(z_b, s) - \chi_S CDF^S_s(s \times w(z_b, s)) \right] dz_b \tag{IA.9}$$

where $f\left(z_b \mid \mu_a, \sigma_a^2\right)$ *is the density function of* z_b *given prior beliefs about CEO ability, and all other objects are defined as above.*

[3] Based on Assumption 1, Figure IA.1 provides a more detailed version of the model timeline (slightly adapting Figure 1). In particular, in period 1 (or 1a and 1b), B and S set their strategies. These strategies are not revealed at this time, but this timing convention defines the notion of the Board's informational advantage. In particular, the Board plays their strategy *after* receiving signal; the shareholder plays their strategy *before*. The assumption of commitment forces S to stick with the strategy that B expects.

Proof. I start with (IA.8). $\mu_{b|z_b}$ and $\sigma^2_{b|z_b}$ are Board beliefs after receiving their signal, hence are known from the perspective of the Board. As $A = \exp(a+\varepsilon_y)$, with a (and beliefs about a) normally distributed, I can write its expectation in terms of means and variances. The probability of vote failure is described in Section 3.3, but as brief overview it is given by the CDF $F^{\mathrm{U}}_{\tilde{z}_s}$, of the unbiased (lognormal) wage of beliefs implied by realizations of $\tilde{z}_s$. The adjustment cost makes the Board's problem dynamic, as they have to factor in the effect of wages on the continuation value.

V^R is value if the CEO retires, so beliefs reset and there is no adjustment cost. In other words, the Board's problem reverts to its $t = 1$ value; it is constant for any state as the prior belief of ability about the CEO talent pool is distributed $N(\mu_0, \sigma_0^2)$ for any state.

The distribution of μ_a' conditional on μ_a and τ is given in Prop. IA.3. However, because the Board has beliefs $\left(\mu_{b|z_b}, \sigma^2_{b|z_b}\right)$, the variance of next period *mean* beliefs (not the variance of beliefs) at the time the Board makes their decision is $\sigma^2_{b|z_b} - \sigma^2_{a'}$. This quantity will be used to take expectation over the continuation value. If the CEO continues, the tenure increases by 1, and the Board must consider the adjustment cost in the next period.

For (IA.9), the objects are the same as the Board's problem, however the shareholders choose s after integrating out z_b. That is, shareholders figure out the Board's wage decision for each z_b, including how they would react to the choice of a particular s, and maximize expected operating income. Under Assumption 2, shareholders do not behave dynamically, and only vote on the current period.

The solution $(w(z_b), s)$ is to be found numerically, each $(w(z_b), s)$ is a best response under commitment (Assumption 1). To sketch the intuition of the solution, fix S' strategy s under commitment. The Board can then back out the probability of failure for each choice of $w(z_b)$, knowing that S must play the threshold. In other words, there is no notion of deviation for the Board. S just needs to maximize (IA.9) for their strategy to be a best response; they cannot deviate at the vote and play a lower threshold. ■

IA1.7. Derivation of Model Statistics

IA1.7.1. Average shareholder value

To interpret the magnitude of the failure-cost parameters, I normalize them by an average level of shareholder value. I define *average shareholder value* as the present value of expected operating profits generated by a CEO of average skill when compensation is chosen to maximize shareholder surplus (i.e., absent SOP considerations and any difference in preferences between the Board and shareholder).

Shareholder-optimal wage. Per-period operating profits in the model are

$$\Pi_t = y_t - \kappa\eta w_t, \qquad y_t = \eta A_t w_t^{\alpha},$$

so $\Pi_t = \eta\big(A_t w_t^{\alpha} - \kappa w_t\big)$. For a CEO of average skill, the shareholder chooses a constant wage $w^\star$ to maximize expected operating profits:

$$w^\star \in \arg\max_{w>0}\ E[A]\, w^{\alpha} - \kappa w.$$

The first-order condition implies

$$w^\star = \left(\frac{\alpha}{\kappa}E[A]\right)^{\frac{1}{1-\alpha}}. \tag{IA.10}$$

Average productivity. Let $\log A$ be normally distributed for a CEO of average skill. Denoting its mean by μ_0 (normalized to zero in estimation) and its variance by σ_A^2, we have

$$E[A] = \exp\big(\mu_0 + \tfrac{1}{2}\sigma_A^2\big). \tag{IA.11}$$

σ_A^2 aggregates the relevant sources of cross-sectional and transitory dispersion in productivity used in the wage/value benchmark (including output risk σ_y^2 and the dispersion implied by private information). As such, the volatility reflects a perceived ex ante average productivity, incorporating that both output and uncertainty are resolved after the Board and Shareholder act on noisy information about CEO ability.[4]

Average shareholder value. Under a stationary benchmark with constant expected operating profits, average shareholder value is the discounted value of profits:

$$\overline{V} \equiv E\left[\sum_{t=0}^{\infty} \delta_B^t \Pi_t\right] = \frac{1}{1-\delta_B} E[\Pi], \tag{IA.12}$$

where $\delta_B \in (0,1)$ is the discount factor. Evaluated at the shareholder-optimal wage $w^\star$, expected per-period operating profits are

$$E[\Pi] = \eta(E[A](w^\star)^\alpha - \kappa w^\star) = \eta(1-\alpha)\, E[A]\,(w^\star)^\alpha,$$

where the last equality uses the first-order condition. Substituting (IA.10)–(IA.11) into (IA.12) yields a closed-form expression for $\overline{V}$ as a function of $(\alpha, \kappa, \eta, \mu_0, \sigma_A^2, \delta_B)$.

Normalizing unscaled structural parameters. I report all unscaled structural parameters in units of average shareholder value. For any unscaled parameter x, I report the normalized quantity $x/\overline{V}$, interpreted as the fraction of average shareholder value associated with the corresponding parameter. In Table 3, standard errors for these normalized parameters are computed via the Delta method, using the estimated variance–covariance matrix of the underlying structural parameters, noting that $\overline{V}$ can be estimated in closed form solely as a function of model parameters.

IA2. Estimation Appendix

IA2.1. Numerical Solution

The model requires 12 estimable parameters, the Board's discount rate and $\mathbf{T}+1$ externally calibrated CEO separation rates. I externally calibrate the Board's discount factor $\delta_B = 0.9615$, which corresponds to annual data. The CEO separation rates are generated by calculating the cross-sectional proportion of CEOs that separate from their firm for a given tenure. I group the remaining 12 parameters as $\mathbb{P}$,

$$\mathbb{P} = \begin{bmatrix}\log\eta & \kappa & \sigma_0 & \sigma_y & \alpha & c_w & \sigma_{z_b} & \sigma_{z_s} & \lambda_0 & \lambda_1 & \chi_B & \chi_S\end{bmatrix} \tag{IA.1}$$

The model's solution proceeds as such

1. Start with a given $\mathbb{P}$
2. Discretize each idiosyncratic shock into an N_z grid; e.g., fix the possible realizations of $\varepsilon_{z_{bt}}$, $\varepsilon_{z_{st}}$, $\varepsilon_{z_{yt}}$, while maintaining the same seed sequence across iterations. I set $N_z = 21$
3. Discretize the state space into a $(N_\mu, \mathbf{T}+1, N_w) = (21, 26, 15)$ grid, call it $\mathcal{S}$, where each tuple (μ_i, τ_j, w_k) indexes current mean belief about CEO ability, tenure (which fully determines beliefs of variance of CEO ability) and the previous wage.
4. Start with a guess of $V_0(\mu, \tau, w)$ as the solution to the static game (i.e., where there is no wage adjustment cost), so V_0 is just the Board's per-period expected utility given optimal choices. Each $V(\mu, \tau, w_{1:N_w})$ starts with the same value.

[4]This corresponds to $E[A] = \exp\left(\frac{1}{2}(\sigma_{0,\text{priv}}^2 + \sigma_y^2)\right)$ with $\sigma_{0,\text{priv}}^2 = \sigma_0^2 + \sigma_{\text{priv}}^2$ and $\sigma_{\text{priv}}^2 = (\sigma_{zb}^2 \sigma_{zs}^2)/(\sigma_{zb}^2 + \sigma_{zs}^2)$.

5. Use Gauss quadrature, interpolation and Prop. IA.2 to estimate the continuation value for each tuple (μ, τ, w)

6. Guess S' voting policy and compute $\mathcal{S}(\mu, \tau, w_{-})$, or S' flow utility given the voting policy

7. Given S' policy, find B's wage policy $\omega(\mu, \tau, w_{-})$ given S' policy using value function iteration on $V(\mu, \tau, w_{-})$

8. Return to 5 and repeat until $\max|V_i - V_{i-1}| < \epsilon = 1\text{e}-5$ and $\max|\mathcal{S}_i - \mathcal{S}_{i-1}| < \epsilon = 1\text{e}-5$

This process returns the Board's wage policy for each element in in the state space and each realization on the grid of ε_{z_b}. Concurrently, it returns the shareholder's policy for each element in the state sapce.

IA2.2. Simulation

I set $N_f = 2000$ firms. Given $a \sim (0, \sigma_0)$, I draw a CEO of skill a for each firm. A CEO spell is the length of time the CEO is matched with a firm. Each period, for each firm, I generate realizations of $\varepsilon_{z_{bt}}$, $\varepsilon_{z_{st}}$ and ε_{yt} (which are drawn under the same sequence of quasi-random numbers each simulation iteration). Given the state, I use the policies described in Section IA2.1 to generate optimal choices. Beliefs update given realizations of $\varepsilon_{z_{bt}}$, $\varepsilon_{z_{st}}$ and ε_{yt}. At the end of each period, for CEOs with tenure $\tau > 0$, they separate (via firing, quitting or retirement) with exogenous probability f_τ.

I generate $N_S = 5$ samples for each simulation. I "fix" randomness across different simulations. That is, each $n_s \in N_S$ sample has the same seed across iterations, only the variance of each CEO ability and each shock changes. Within each simulation n_s, I allow 20 years of burn-on data, and use 15 years as my analysis sample.

IA2.3. Estimation

I estimate the 12 parameters in (IA.1) As mentioned above, the Board's discount factor δ_B is calibrated to 0.9615, and separation rates are calibrated to match observed separation rates in the sample. I estimate the remaining model parameters by finding a vector $\mathbb{P}$ of parameters that minimizes the weighted distance between simulated and data moments. That is, given model moment $m(\mathbb{P})$ and data moments $m(X)$ and an appropriate weighting matrix W, I minimize

$$\min_{\mathbb{P}} \Big[m(\mathbb{P}) - m(X)\Big]' W \Big[m(\mathbb{P}) - m(X)\Big] \tag{IA.2}$$

Weight matrix. I use the inverse of the variance-covariance matrix of the data moments as the weight matrix in estimation: $W = V^{-1}$, which is the efficient weight matrix for SMM. To produce V, I use the influence function (IF) approach. I cluster V by firm-CEO spell, which is the appropriate clustering approach given that the model's learning and shock processes are all confined within CEO spell. Formally, for each moment m_k, let IF_{ik} be observation i's marginal effect on the estimated m_k. Then

$$\sqrt{N}(\hat{m} - m_0) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} IF_i + o_p(1)$$

and $\hat{V}$ can be consistently estimated via

$$\hat{V} = \frac{1}{N} \sum_{g=1}^{G} \left(\sum_{i \in g} IF_i \right) \left(\sum_{i \in g} IF_i \right)',$$

where g is a firm-CEO spell. The weight matrix used in the estimation is $\hat{W} = \hat{V}^{-1}$.

IA2.4. Optimization algorithm

My goal is to find the global minimum of the SMM objective function described in Section IA2.3. To leverage the efficiency of parallel computing, I use a somewhat modified version of the TikTak global optimization algorithm described in Arnoud et al. (2019).[5] The modifications are designed to take advantage of high performance computing to minimize computing time. The global optimization routine can be described as such:

1. **Parallel local minimization**

   i. Generate bounds for each parameter. This is a holistic step, yet the bounds should be narrow enough to allow for the subsequent quasi-random sequences to adequately cove the space, but wide enough so that I maximize the chance of finding the global minimum.

   ii. Using the bounds, generate a Sobol sequence of length N. Sobol points are quasi-random points which are intended to mimic a draw from from a uniform distribution. In my setup, I set $N = 50,000$ or $N = 100,000$.

   iii. For each $n \in N$ of the Sobol points, evaluate the objective function at n. Keep N_p of the points with the smallest initial function value. I set $N_p = 100$ or $N_p = 200$.

   iv. For each $p \in N_p$, solve for the local minimum. In my setup, I use Nelder-Mead locally, which gives N_p "promising" candidates for the global minimum.

2. **Parallel global minimization**. This step slightly modifies the TikTak routine to take advantage of parallel computing.

   i. Take the $p \in N_p$ candidates for the global minimum from above and sort in ascending order. Set $i = 1$, so the best minimum so far is indexed by i.

   ii. Take the best minimum so far, labeled p_i^*. Generate $N_p - i$ convex combinations using the TikTak methodology. That is, for $j \in N_p - i$, $p_{ji}^{cand} = \theta_{ji} p_i^* + (1 - \theta_{ji}) p_j$, where $\theta_{ji} \in [0, 1]$ and approaches 1 as j increases.

   iii. Compute the local minimum of each $N_p - i$ point in parallel. If p_i^* is the best, then exit the routine and p_i^* is the candidate global minimum. Else,

   iv. For the first j such that function value of p_{ji}^{cand} is less than that of p_i^*, stop all subsequent (unfinished) local minimization routines for $j' \in N_p - i$, and $j' > j$. Update i += p and return to ii.

   This routine will return p_i^* as the global minimum.

3. **Polish global minimum**. Using stricter stopping criteria and a large number of function iterations, polish the global minimum p_i^* using a local minimization routine, e.g., Nelder-Mead.

[5] I modified code from https://github.com/tpapp/MultistartOptimization.jl, which is based upon the original TikTak code: https://github.com/serdarozkan/TikTak. See also Liu (2022) for a recent example.

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